

A Unified Psychoacoustic–Semantic Representation for Speech Emotion Recognition, and the Evaluation Protocol Required to Assess It

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Abstract

Speech emotion recognition increasingly relies on self-supervised representations, yet whether perceptually motivated psychoacoustic descriptors remain complementary to them is unresolved, and published eight-class accuracies on the RAVDESS corpus range from 39.6% to 99.0%. We propose PAER-LLM, a five-layer framework that unifies 34 psychoacoustic descriptors with frozen HuBERT embeddings through a learned per-dimension gate and conditions an instruction-tuned language model on the resulting posterior. We formalise the unified-representation problem, the protocol-dependence of its assessment, and the resolution at which a representational claim is admissible. Evaluating one implementation under six protocols that differ only in partitioning and model selection, with architecture, features, optimiser and schedule held fixed, accuracy falls from 94.79% under a contaminated random split to $64.28\% \pm 5.30$ under speaker-disjoint five-fold cross-validation—a spread of 30.5 points. Under the most defensible protocol the largest difference between three fusion operators is 2.03 points, below the design’s 3.47-point minimum detectable effect. Contamination also conceals a 21.6-point gender disparity, shrinking it to 2.5 points. We prove that single-token cross-modal attention is analytically degenerate, its query stream receiving exactly zero gradient, and show that a corrected multi-token form is outperformed by a single gate at $3.1\times$ the parameters.

Index Terms

Speech emotion recognition, psychoacoustic features, representation learning, feature fusion, evaluation protocol, data leakage, speaker-independent evaluation, large language models.

I. INTRODUCTION

Speech emotion recognition (SER) underpins affective human–computer interaction, and is supported by large self-supervised speech encoders and an extensive benchmark literature [1]–[3]. This paper is concerned with two problems that the field treats separately and that, we argue, cannot be separated.

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A. Problem Statement

Emotion in speech is carried by perceptually salient acoustic properties. Listeners recover a talker’s affective state from fundamental frequency and its movement, from intensity, and from how energy is distributed across the auditory critical bands—quantities that hearing science has described for six decades and that standardised parameter sets such as GeMAPS/eGeMAPS formalise [4], [5]. Modern SER, however, is built almost entirely on self-supervised transformer encoders trained on raw waveforms with no perceptual objective [6]–[8]. These two accounts of what matters in an emotional utterance have largely stopped talking to each other. The first problem this paper addresses is therefore representational: *do perceptually motivated psychoacoustic descriptors still carry emotional information that a large self-supervised encoder does not already provide, and if so, how should the two be combined into a single representation?* The question is not settled. Dixit *et al.* [9] regress self-supervised embeddings onto eGeMAPS categories and find that deep representations recover only part of the handcrafted perceptual space, which implies complementarity; yet systems that add handcrafted descriptors to self-supervised embeddings report gains of one to three accuracy points, which is small enough that it may not be a gain at all.

That last clause is the second problem, and it turns out to govern the first. Published eight-class accuracies on the Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS) [10]—the corpus on which such comparisons are most often made—span 39.6% [11] to 99.0% [12] within the last two years. They cannot both characterise the same task. The discrepancy is not principally architectural: it follows from how the corpus is partitioned. RAVDESS comprises 1440 speech utterances produced by 24 professional actors, each reading the same two sentences across all eight emotion categories at two intensities with two repetitions. A random utterance-level partition therefore guarantees that for every test item there exists a training item from the same speaker, with the same lexical content, at the same emotional intensity, differing only in take. A classifier can succeed by recognising the talker rather than the emotion. When the confounding effect of the evaluation design exceeds the effect being measured, no representational question can be answered on that evidence, however carefully the representation is built.

The two problems compound. A researcher proposing a unified psychoacoustic and self-supervised representation, as we do, has no way to establish that the psychoacoustic stream earns its place if the measurement apparatus cannot resolve differences of the size that stream is likely to produce. Conversely, a field that reports architectural improvements of one to three points under undisclosed partitioning schemes accumulates a literature in which representational claims cannot be checked. This paper therefore treats the two as one problem. We formalise the unified representation and the protocol that assesses it in a single framework (Section III), construct the representation (Section IV), state precisely how much evidence would be required to accept it, and then measure how much evidence the field’s standard protocols actually supply. The answer is: considerably less than is required, and the shortfall is large enough to swallow the entire architectural literature on this corpus.

B. Research Objective and Scope

The objective of this work is to propose and characterise a unified representation framework that integrates psychoacoustic emotional features with self-supervised speech representations, and to condition a large language model on the emotional evidence that framework produces. We name the framework PAER-LLM (Psychoacoustically-Aware Emotion Recognition with a Language Model) and specify it as five layers in Section IV.

Three boundaries on that objective should be stated at the outset.

First, the framework is *unimodal*. Both streams are computed from one audio signal; they are two feature views of the same modality, not two modalities. We therefore use “unified representation” and never “multimodal”, and distinguish our setting from audio-visual work explicitly.

Second, Layers 1–3 are evaluated here. Layer 4, which conditions an instruction-tuned language model on the classifier’s posterior, is specified and implemented but *not* evaluated: we report no human study, no automatic metric and no ablation for it. Layer 5 is a demonstration interface. We mark this division in Fig. 1 rather than leaving it to be inferred, and Section VIII-E sets out the calibration experiment that would make Layer 4 load-bearing.

Third, the empirical scope is a single corpus. All numerical results are on RAVDESS. The analytical results of Section III-F are corpus-independent; the magnitudes are not.

We stress one point of interpretation. The configuration of PAER-LLM that we adopt is statistically indistinguishable from plain concatenation of the two encoded streams, and its originally proposed fusion mechanism does not work at all, in a sense we prove in Proposition 1. We report both against our own interest, because the alternative—selecting the configuration that flatters the proposed method—is precisely the researcher degree of freedom that Nießl *et al.* [13] identify as a source of systematic over-optimism.

C. Contributions

- 1) **A formal problem formulation** (Section III) that states the unified-representation objective, the information-theoretic condition under which the psychoacoustic view can contribute, the evaluation protocol as an operator on the corpus, and an *admissibility* criterion that a representational claim must satisfy to be supported by a given design.
- 2) **A unified psychoacoustic–semantic representation framework.** PAER-LLM encodes 34 perceptually motivated descriptors and a frozen mean-pooled HuBERT-base embedding into a common 128-dimensional space and combines them with a learned per-dimension gate, whose posterior conditions an instruction-tuned language model (Section IV, Algorithms 3–4). All three fusion operators compared emit exactly 128 dimensions, so downstream capacity is fixed and the comparison isolates the operator.
- 3) **Three analytical properties of pooled-vector fusion** (Section III-F): single-token cross-modal attention is exactly degenerate and gradient-inert; gated fusion is confined to the per-coordinate convex hull of its inputs while concatenation is not, so neither operator contains the other; and the

parameter and multiply—accumulate cost of token cross-attention exceeds both by $3.1\times$ and $5.3\times$ respectively.

- 4) **A controlled decomposition of the evaluation-protocol effect** (Section VII-A, Algorithm 5), isolating duplicate records, speaker-dependent partitioning, test-set model selection and single-split reporting within one fixed system. To our knowledge this is the first study to attribute a RAVDESS performance gap to its individual protocol components rather than to report two endpoints.
- 5) **A protocol-to-architecture effect ratio and a stated resolution limit.** Under the most defensible protocol the largest difference between three fusion operators is 2.03 points against a 30.5-point protocol effect; the design’s minimum detectable effect is 3.47 points, so the entire architectural range is inadmissible under our own criterion (Sections VII-B and VII-D).
- 6) **Evidence that contamination conceals fairness failures.** Under the contaminated protocol the female–male accuracy disparity is 2.5 points; under speaker-disjoint evaluation of the same system it is 21.6 points, and the dominant per-class failure (*happy*, F_1 0.511) is invisible (F_1 0.915). Protocol choice therefore distorts not only reported accuracy but the fairness and error audits built on it (Section VII-F).
- 7) **A reproducible evaluation harness** with published fold assignments, the contaminated-protocol reproduction, a corpus-integrity check, the complementarity and calibration analysis, and per-fold prediction files.

D. Organisation

Section II reviews the relevant literature including recent systems. Section III formalises the problem. Section IV specifies the framework. Section V sets out the research methodology and the benchmarking design. Section VI gives the experimental configuration. Section VII reports results, including the performance matrix and complexity analysis. Section VIII discusses, Section IX states limitations, and Section X concludes.

II. RELATED WORK

A. Self-Supervised Representations for Emotion Recognition

Self-supervised encoders—wav2vec 2.0 [6], HuBERT [7] and WavLM [8]—underpin most competitive SER systems, and SUPERB [14] established frozen-encoder probing as a standard evaluation. Pepino *et al.* [15] showed that a learned weighted combination of intermediate layers substantially outperforms the final layer for emotion, consistent with layer-wise probing analyses showing an acoustic–phonetic–acoustic trajectory across depth [16], [17]. Chiu *et al.* [18] report that speaker-state and prosodic attributes are most robustly represented in middle layers while final layers suppress speaker identity. Wagner *et al.* [19] provide the most systematic study of transformer fine-tuning for dimensional SER, including robustness and fairness analyses. Emotion-specialised pretraining extends this line [20], [21].

We use frozen mean-pooled HuBERT-base representations. This is deliberately conservative: it isolates the contribution of the psychoacoustic stream and the fusion operator from that of encoder adaptation, which is known to be large [22].

B. Psychoacoustic and Prosodic Representations

That emotion is carried by perceptually salient acoustic correlates remains supported by recent systematic review [23], [24]. Standardised parameter sets such as GeMAPS/eGeMAPS [4], extracted via openSMILE [25], formalise this premise. The critical-band concept and its analytical approximation are due to Zwicker [5] and Zwicker and Terhardt [26] respectively. Whether such descriptors retain value alongside self-supervised embeddings is the representational question this framework is built to test [9]. Perceptually motivated spectral front-ends have shown advantages over mel filterbanks, particularly under noise [27], [28].

C. Fusion of Heterogeneous Feature Streams

Cross-attention is the default mechanism for fusing heterogeneous representations in affective computing [29]–[31], including systems fusing handcrafted acoustic descriptors with self-supervised embeddings [32]. Gated variants are common [33], and Nasersharif and Azad [34] compare attention-based fusion against simpler classifier combination. Much of this literature is genuinely multimodal—audio with video or text—whereas our two streams are views of one signal; the distinction matters because cross-modal attention is motivated by alignment between sequences that our pooled setting does not provide (Section VIII-C).

A parallel literature questions how much attention contributes. Dong *et al.* [35] prove that pure attention loses rank doubly exponentially with depth; Zhai *et al.* [36] characterise pathological attention-entropy regimes; and Bibal *et al.* [37] survey whether attention weights constitute evidence of learned interaction. Proposition 1 is related but distinct: rank collapse is asymptotic in depth, whereas our degeneracy is *exact* and arises from sequence length one, independently of depth, head count or scaling.

D. Language Models for Emotion-Sensitive Interaction

Emotionally aware dialogue has been studied through emotional support conversation [38], empathetic response generation [39], and benchmarks of emotional competence [40]. Architecturally, systems divide into end-to-end audio-conditioned models [41]–[43] and cascaded designs. Recent work shows a frozen LLM can be made to perceive paralinguistic style through encoder alignment [44], and that structured, interpretable speech descriptors can condition an LLM for explainable emotion recognition [45]. We adopt a cascaded design conditioned on the classifier’s full posterior rather than its top-1 label, informed by work on eliciting calibrated confidence [46] and on principled abstention [47].

E. Evaluation Protocol, Leakage and Reproducibility

Kapoor and Narayanan [48] document leakage-driven irreproducibility across seventeen fields. Elan-govan *et al.* [49] quantify train–test overlap in NLP benchmarks and show that near-duplicate examples permit high scores through memorisation. Rosenblatt *et al.* [50] find that repeated subjects across partitions inflate performance most severely, with small datasets most vulnerable—a description that fits RAVDESS closely. Gorman and Bedrick [51] show that conclusions drawn from standard splits frequently fail to

replicate under resampling, and Bouthillier *et al.* [52] demonstrate that ignoring sources of benchmark variance inflates apparent significance [13], [53].

The standard we advocate is not new. The INTERSPEECH 2009 Emotion Challenge [54] established speaker-independent partitioning as the norm nearly two decades ago, and the Interspeech 2025 challenge on naturalistic SER [55] retains it. EmoBox [56] now distributes standardised speaker-independent splits across corpora, including RAVDESS. What has happened since 2009 is erosion rather than absence.

F. Recent Systems and the State of the Art

Because our central claim concerns comparability rather than rank, we survey recent RAVDESS systems by *protocol* rather than by score.

Among speaker-dependent designs, Jafarzadeh *et al.* [12] report 97.8–99.0% for self-supervised encoders with a lightweight classifier under an explicitly random stratified 80/20 split; emotion2vec [21] reports 82.43% under what it calls “random leave-one-out 10-fold CV”, a phrase that reads as speaker-independent but denotes a random utterance-level partition; and Ibrahim *et al.* [57] report 95.83% for audio-visual hybrid fusion.

Among speaker-independent designs, CAMuLeNet [11] reports 82.3% under ten-fold leave-speaker-out and, valuably, also reports single-encoder baselines under the identical protocol—58.7% for HuBERT, 65.4% for WavLM, 39.6% for wav2vec 2.0. Luna-Jiménez *et al.* [58] report 81.82% under subject-wise five-fold cross-validation with published fold assignments. Portal *et al.* [59] re-validate state-of-the-art SER transformers under strictly speaker-independent conditions and find most single-corpus results below 40%. Ibrahim *et al.* [57] report both protocols for one system: 95.83% random against $48.06\% \pm 9.76$ leave-one-speaker-out.

Two observations follow. First, the two most informative recent papers are those that report *both* protocols or several encoders under *one* protocol; single-number reports are the least usable. Second, the speaker-independent class spans 44.7 points, wider than the speaker-dependent class, so protocol annotation alone does not make figures comparable. Our contribution to this literature is a controlled decomposition: rather than comparing across papers that differ in many respects, we vary the protocol alone within one system and attribute the resulting gap to its components.

III. PROBLEM FORMULATION

A. Notation and Task

Let $\mathcal{A}$ denote the space of speech waveforms and $\mathcal{Y} = \{1, \dots, C\}$ the set of emotion categories, $C = 8$. A corpus is a set $\mathcal{D} = \{(a_k, y_k, s_k)\}_{k=1}^N$ of utterances $a_k \in \mathcal{A}$ with labels $y_k \in \mathcal{Y}$ and speaker identifiers $s_k \in \mathcal{S}$; for RAVDESS, $N = 1440$ and $|\mathcal{S}| = 24$. Two feature maps extract complementary views of the same signal,

$$\varphi_p : \mathcal{A} \rightarrow \mathbb{R}^{d_p}, \quad \varphi_h : \mathcal{A} \rightarrow \mathbb{R}^{d_h}, \quad (1)$$

with $d_p = 34$ perceptually motivated descriptors and $d_h = 768$ dimensions from a frozen self-supervised encoder. Write $\mathbf{x}_p = \varphi_p(a)$ and $\mathbf{x}_h = \varphi_h(a)$.

B. The Unified Representation Problem

We seek stream encoders $\text{Enc}_p : \mathbb{R}^{d_p} \rightarrow \mathbb{R}^d$ and $\text{Enc}_h : \mathbb{R}^{d_h} \rightarrow \mathbb{R}^d$, a fusion operator $\omega : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ drawn from a family Ω , and a classifier $g_\psi : \mathbb{R}^d \rightarrow \Delta^{C-1}$, such that the *unified representation*

$$\mathbf{z} = \omega(\text{Enc}_p(\mathbf{x}_p), \text{Enc}_h(\mathbf{x}_h)) \in \mathbb{R}^d \quad (2)$$

minimises the expected risk

$$(\theta^*, \omega^*) = \arg \min_{\theta, \omega \in \Omega} \mathbb{E}_{(a,y) \sim P} [\ell(g_\psi(\mathbf{z}), y)], \quad (3)$$

subject to the *capacity constraint*

$$\dim(\omega(\cdot, \cdot)) = d \quad \text{for every } \omega \in \Omega. \quad (4)$$

Constraint (4) is what makes an ablation over Ω interpretable: every operator hands the classifier the same number of dimensions, so a difference in risk is attributable to the operator and not to downstream width. We take $\Omega = \{\text{CONCAT}, \text{GATE}, \text{XATTN}\}$, defined in Section IV-C.

C. When Can the Psychoacoustic View Contribute?

Objective 1 asks whether φ_p is worth including at all. Let A be a random utterance with label Y . The relevant quantity is the *unique information* the perceptual view carries about the label given the self-supervised view,

$$U_p = I(\varphi_p(A); Y \mid \varphi_h(A)) \geq 0, \quad (5)$$

so that

$$I(\varphi_p(A), \varphi_h(A); Y) = I(\varphi_h(A); Y) + U_p. \quad (6)$$

Complementarity is exactly the condition $U_p > 0$; *redundancy* is $U_p = 0$. Two consequences shape the rest of the paper.

First, $U_p > 0$ is necessary but not sufficient for an accuracy gain: a finite estimator trained on $N = 1440$ utterances may fail to extract information that is present. A null result on accuracy therefore does not establish redundancy.

Second, $U_p > 0$ implies that, in the population limit, there exist utterances the perceptual view classifies correctly and the self-supervised view does not. Writing E_m for the error set of the system trained on view m , this motivates measuring $|E_h \setminus E_p|$ directly rather than inferring complementarity from a difference of means—a distinction that becomes decisive once Section VII-D bounds what the difference of means can resolve. Algorithm 6 implements the measurement.

D. Evaluation Protocol as an Operator

A *protocol* is a partition operator

$$\Pi : (\mathcal{D}, s) \longmapsto (I_{\text{tr}}, I_{\text{va}}, I_{\text{te}}), \quad (7)$$

mapping a corpus and a seed to disjoint index sets, together with a rule naming the partition on which the model is selected. The reported risk is the protocol-conditioned estimate

$$\widehat{R}_{\Pi}(f) = \frac{1}{|I_{\text{te}}|} \sum_{k \in I_{\text{te}}} \mathbb{K}[f(a_k) \neq y_k]. \quad (8)$$

Definition 1 (Speaker-disjointness). Π is *speaker-disjoint* if $\mathcal{S}(I_{\text{tr}}) \cap \mathcal{S}(I_{\text{te}}) = \emptyset$, where $\mathcal{S}(I) = \{s_k : k \in I\}$.

Two effects can now be compared on the same scale. For a fixed architecture $\mathcal{A}$ and two protocols, the *protocol effect* is

$$\Delta_{\Pi}(\mathcal{A}) = |\widehat{R}_{\Pi_1}(f_{\mathcal{A}}) - \widehat{R}_{\Pi_2}(f_{\mathcal{A}})|; \quad (9)$$

for a fixed protocol, the *architecture effect* is

$$\Delta_{\Omega}(\Pi) = \max_{\omega, \omega' \in \Omega} |\widehat{R}_{\Pi}(f_{\omega}) - \widehat{R}_{\Pi}(f_{\omega'})|. \quad (10)$$

The paper’s central empirical claim is that $\Delta_{\Pi}(\mathcal{A}) \gg \Delta_{\Omega}(\Pi)$ on this corpus, by a factor of roughly fifteen. Because $\Delta_{\Omega}(\Pi)$ is small, the choice of which fixed $\mathcal{A}$ anchors the ladder in (9) shifts the measured protocol effect by at most $\Delta_{\Omega}(\Pi)$, which is 2.03 points here.

E. Admissibility and Design Sensitivity

Let δ be a claimed improvement measured as a mean paired difference over n folds with standard deviation σ_{Δ} . For a two-sided paired t -test at level α and power $1 - \beta$, the *minimum detectable effect* is

$$\delta_{\min}(n) = \sigma_{\Delta} \cdot d_z^*(n, \alpha, \beta), \quad (11)$$

where d_z^* solves $1 - \beta = \Pr[|T'_{n-1}(d_z \sqrt{n})| > t_{n-1, 1-\alpha/2}]$ and T' is the noncentral t distribution.

Definition 2 (Admissibility). A *representational or architectural claim* of size δ , made on corpus $\mathcal{D}$ under protocol Π with n folds, is *admissible* only if $|\delta| \geq \delta_{\min}(n)$.

This is a statement about design, not about an observed p -value: post-hoc power computed from an observed effect is a monotone function of that effect’s p -value and carries no additional information [60]. Section VII-D computes $\delta_{\min}$ for our design and applies Definition 2 to our own results, which it fails.

F. Properties of Pooled-Vector Fusion

The three operators in Ω admit analytical comparison, and two of the three results below were not anticipated when the framework was designed.

Proposition 1 (Degeneracy of single-token cross-attention). *Let $\mathbf{q}, \mathbf{k}, \mathbf{v} \in \mathbb{R}^{1 \times d_k}$ be the projected query, key and value of a scaled dot-product attention block whose query and key/value sequences each have length one. Then the attention output satisfies $\mathbf{A} = \mathbf{v}$ identically, and $\partial \mathbf{A} / \partial \mathbf{q} = \mathbf{0}$, for any number of heads, any head dimension and any scaling.*

Proof: The score matrix is $\mathbf{S} = \mathbf{q}\mathbf{k}^\top / \sqrt{d_k} \in \mathbb{R}^{1 \times 1}$, and $\text{softmax}(\mathbf{S})_{11} = e^{S_{11}} / e^{S_{11}} = 1$ for every $S_{11} \in \mathbb{R}$. Hence $\mathbf{A} = 1 \cdot \mathbf{v}$, which does not depend on $\mathbf{q}$; the partial derivative is therefore identically zero. Multiple heads apply the same argument per head, and any scaling factor cancels in the ratio. ■

Corollary 1. *If $\omega = \text{XATTN}$ with $K = 1$ and the psychoacoustic stream supplies the query, then $\partial \mathbf{z} / \partial \mathbf{e}_p = \mathbf{0}$: the psychoacoustic encoder receives no gradient and its parameters remain at initialisation. The model is exactly the unimodal self-supervised model plus an unused parameter block, and cannot in principle demonstrate $U_p > 0$.*

Corollary 1 is the reason $K > 1$ is asserted at line 11 of Algorithm 3; Section VII-G confirms it empirically.

Proposition 2 (Convex-hull confinement of gated fusion). *For $\mathbf{z} = \mathbf{g} \odot \mathbf{e}_p + (\mathbf{1} - \mathbf{g}) \odot \mathbf{e}_h$ with $\mathbf{g} \in (0, 1)^d$,*

$$\min(e_{p,i}, e_{h,i}) < z_i < \max(e_{p,i}, e_{h,i}) \quad \forall i, \quad (12)$$

hence $\|\mathbf{z}\|_\infty \leq \max(\|\mathbf{e}_p\|_\infty, \|\mathbf{e}_h\|_\infty)$. No such bound holds for $\mathbf{z} = \mathbf{W}_c[\mathbf{e}_p; \mathbf{e}_h]$. Consequently neither operator’s function class contains the other.

Proof: Each z_i is a strict convex combination of $e_{p,i}$ and $e_{h,i}$ because $g_i \in (0, 1)$, giving the bound. For the converse, take $\mathbf{W}_c$ with $\mathbf{z} = 2(\mathbf{e}_p + \mathbf{e}_h)$, whose ∞ -norm exceeds the bound whenever $\mathbf{e}_p, \mathbf{e}_h \neq \mathbf{0}$ share a sign in some coordinate; and note that $\mathbf{g}$ depends nonlinearly on $[\mathbf{e}_p; \mathbf{e}_h]$, so the gated map is not linear and is not realisable by any fixed $\mathbf{W}_c$. ■

Proposition 2 explains the empirical near-tie of Section VII-B without appealing to noise alone: the two operators span overlapping but distinct function classes, and the emotion-relevant signal in a pair of pooled vectors evidently lies in the intersection. It also identifies what the gate buys that concatenation does not—an explicit per-coordinate attribution $\mathbf{g}$ between the perceptual and semantic views, which Section VII-C analyses.

Proposition 3 (Cost ordering). *With encoder width d , token count K , and unshared expansions and projections in both attention directions, the trainable fusion parameter counts are $2d^2 + d$ for both CONCAT and GATE, and $4(Kd^2 + Kd) + 2(4d^2 + 4d) + 4d + (2d^2 + d)$ for XATTN. At $d = 128$, $K = 4$ this gives 32,896 against 429,696, and total trainable counts of 187,656 against 584,456, a ratio of $3.11\times$; the corresponding multiply-accumulate ratio per utterance is $5.28\times$ (Table XII).*

Reading Fig. 1. The figure is read top to bottom as a pipeline and left to right within each layer. The left tab names the layer; the right panel states what that layer is responsible for delivering. Layer 1 ends in a verified corpus with duplication factor $r = 1$, which is the precondition for every number in this paper. Layer 2 shows the two views side by side with their dimensionalities, and the “+” between them denotes that they are carried forward jointly rather than combined at this stage. Layer 3 is the paper’s object of study: the two encoders map 34 and 768 dimensions to a common 128, the gate combines them, and the classifier emits an eight-way posterior; the italic line beneath records the two operators evaluated and not

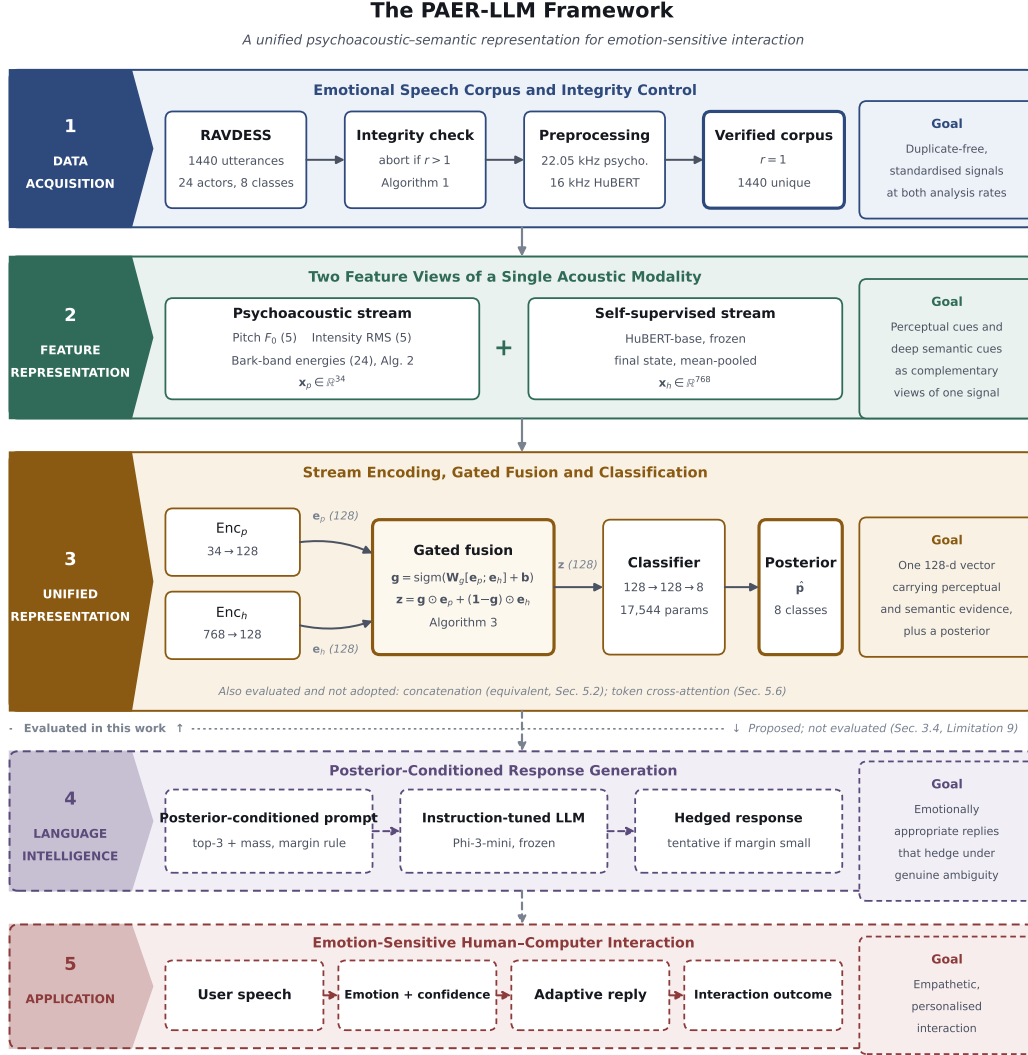


Fig. 1. The PAER-LLM framework. Two feature views of one audio signal—34 psychoacoustic descriptors and a 768-dimensional frozen HuBERT-base embedding—are independently encoded to 128 dimensions and combined by a learned per-dimension gate into a single unified representation $\mathbf{z} \in \mathbb{R}^{128}$. Two alternative operators, concatenation and token cross-attention, are evaluated and not adopted. The resulting posterior over eight categories conditions an instruction-tuned language model; only the posterior is passed, and learned activations are not. Layers 4 and 5 are drawn dashed, below the divider, because they are proposed and implemented but not evaluated in this work. Edge annotations give tensor shapes.

adopted. The dashed divider below Layer 3 is the boundary of the experimental evidence—everything above it is measured, everything below it is specified and implemented but not evaluated, and the dashed outlines of Layers 4 and 5 carry that distinction visually rather than relying on the reader to recall it from the text.

IV. THE PAER-LLM FRAMEWORK

Fig. 1 summarises the whole; we describe each layer in turn.

Algorithm 1 Corpus integrity verification**Require:** dataset root D ; expected corpus size N_{exp} **Ensure:** unique utterance set U ; duplication factor r

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1:  $P \leftarrow \text{SORT}(\text{all } .\text{wav paths under } D)$ 
2:  $U \leftarrow \emptyset$ ;  $\mathcal{S} \leftarrow \emptyset$ 
3: for all  $p \in P$  do
4:    $b \leftarrow \text{BASENAME}(p)$ 
5:   if  $b \notin \mathcal{S}$  then
6:      $\mathcal{S} \leftarrow \mathcal{S} \cup \{b\}$ ;  $U \leftarrow U \cup \{p\}$ 
7:   end if
8: end for
9:  $r \leftarrow |P|/|U|$ 
10: if  $r > 1$  then
11:   abort unless duplication is explicitly requested
12: end if
13: if  $|U| \neq N_{\text{exp}}$  then
14:   warn
15: end if
16: return  $U, r$ 

```

$\triangleright r > 1$: exact copies of test items enter training

A. Data Acquisition Layer

The corpus is RAVDESS [10], restricted to speech: 1440 utterances from 24 actors (12 female, 12 male) across eight categories. All categories except *neutral* contain 192 utterances; *neutral* contains 96, as it is recorded at a single intensity. The corpus also contains 1012 song recordings, which we exclude. Audio is resampled to 22,050 Hz for psychoacoustic analysis and to 16,000 Hz for the self-supervised encoder, matching its pretraining condition.

Corpus integrity. We recommend that this layer include an explicit integrity check. In our case the distributed archive had been unpacked twice into nested directories, so a recursive file walk returned 2880 paths for 1440 distinct recordings and every derived feature table contained each utterance exactly twice. Because the duplicate rows are byte-identical, no summary statistic reveals the problem: class balance, feature distributions and file counts all appear plausible. The defect surfaces only when the number of unique filenames is compared against the number of rows. We report it because it is silent, easy to introduce, and materially inflates every downstream result. The released harness performs the check and refuses to proceed on a mismatch (Algorithm 1), returning the duplication factor r so that the contaminated condition can be reconstructed deliberately for protocol P1.

B. Feature Representation Layer

Psychoacoustic stream. We extract 34 perceptually motivated descriptors per utterance. *Pitch* (5): mean, maximum, minimum, standard deviation and range of voiced F_0 , from Praat’s autocorrelation tracker over 75–600 Hz. No utterance is wholly unvoiced, so the unvoiced fallback is never exercised. *Intensity* (5):

Algorithm 2 Nyquist-adaptive Bark filter bank**Require:** sample rate f_s ; FFT size n ; band count $B = 24$ **Ensure:** filter bank $\Phi \in \mathbb{R}^{B \times (1+n/2)}$

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1:  $\mathbf{f} \leftarrow$  FFT bin centres on  $[0, f_s/2]$ ;  $\mathbf{z} \leftarrow z(\mathbf{f})$  ▷ Eq. (13)
2:  $z_{\max} \leftarrow z(f_s/2)$  ▷ adaptive; canonical scheme reaches 24.0 at 15.5 kHz
3:  $\mathbf{e} \leftarrow \text{Linspace}(0, z_{\max}, B + 2)$ 
4: for  $i = 1$  to  $B$  do
5:    $(\ell, c, u) \leftarrow (e_i, e_{i+1}, e_{i+2})$ 
6:    $\Phi_{i,:} \leftarrow \max(0, \min(\frac{\mathbf{z}-\ell}{c-\ell}, \frac{u-\mathbf{z}}{u-c}))$ 
7:    $\Phi_{i,:} \leftarrow \Phi_{i,:} / \sum_j \Phi_{i,j}$  ▷ unit area
8: end for
9: verify  $\text{Var}_k(\Phi_{i,:} \mathbf{S}_k) > 0 \forall i$ 
10: return  $\Phi$ 

```

the same five statistics of frame-level RMS energy. We call these *intensity* and not *loudness*: loudness in the psychoacoustic sense [61] involves critical-band excitation, masking and a compressive sone scale, none of which RMS energy models. *Bark-band energies* (24): mean energy in each of 24 bands on the critical-band-rate scale of Zwicker and Terhardt [26],

$$z(f) = 13 \arctan(0.00076f) + 3.5 \arctan\left(\left(\frac{f}{7500}\right)^2\right), \quad (13)$$

where f is in Hz and z in Bark. Note that the argument of the second arctangent is squared, not the arctangent itself. Triangular filters are area-normalised so band energy is not biased by the increasing bandwidth of higher bands.

One deviation requires comment. The 24 Zwicker bands extend to 15.5 kHz ($z = 24.0$), above the 11.025 kHz Nyquist frequency of our 22.05 kHz analysis rate. Rather than retain bands that would be identically zero, we place 24 equal-width bands over $[0, z(f_{\text{Nyq}})] = [0, 22.86]$ Bark, so all lie below Nyquist by construction. The cost is that the upper 1.14 Bark of the audible range is unrepresented and the bands are not the canonical Zwicker bands. Since RAVDESS is distributed at 48 kHz, analysing at ≥ 32 kHz would remove the constraint entirely; we regard this as a defect of the present configuration rather than a design choice to defend. Algorithm 2 gives the construction.

Self-supervised stream. HuBERT-base [7], pretrained on LibriSpeech-960 and held frozen. The final hidden state is mean-pooled over time to give a 768-dimensional utterance embedding.

C. Unified Representation Layer

Each stream is encoded by an independent network $\mathbf{e}_m = \text{Enc}_m(\mathbf{x}_m) \in \mathbb{R}^d$, $m \in \{p, h\}$, with $d = 128$ and the stack $\text{Linear}(d_m \rightarrow d) \rightarrow \text{LayerNorm} \rightarrow \text{GELU} \rightarrow \text{Dropout}(0.3) \rightarrow \text{Linear}(d \rightarrow d) \rightarrow \text{LayerNorm} \rightarrow \text{GELU}$, using post-normalisation, PyTorch default initialisation and no residual connection [62]–[64]. Encoding both streams to a common width before fusion is what satisfies the capacity constraint (4). We compare three operators.

Concatenation. $\mathbf{z} = \mathbf{W}_c[\mathbf{e}_p; \mathbf{e}_h]$ with $\mathbf{W}_c \in \mathbb{R}^{2d \times d}$.

Gated fusion (adopted).

$$\mathbf{g} = \text{sigm}(\mathbf{W}_g[\mathbf{e}_p; \mathbf{e}_h] + \mathbf{b}), \quad \mathbf{z} = \mathbf{g} \odot \mathbf{e}_p + (\mathbf{1} - \mathbf{g}) \odot \mathbf{e}_h. \quad (14)$$

Each coordinate of $\mathbf{g}$ is the weight assigned to the perceptual view at that coordinate, so $\mathbf{g}$ is a directly readable per-dimension allocation between the two representations—a property neither alternative has (Proposition 2).

Token cross-attention. Each pooled vector is expanded into $K = 4$ learned tokens,

$$\mathbf{T}_m = \text{reshape}(\mathbf{W}_m \mathbf{e}_m) \in \mathbb{R}^{K \times d}, \quad m \in \{p, h\},$$

and the streams attend bidirectionally,

$$\mathbf{A}_{p \rightarrow h} = \text{MHA}(\mathbf{T}_p, \mathbf{T}_h, \mathbf{T}_h), \quad \mathbf{A}_{h \rightarrow p} = \text{MHA}(\mathbf{T}_h, \mathbf{T}_p, \mathbf{T}_p), \quad (15)$$

with four heads of dimension 32 and attention dropout 0.3. The expansions and the attention projections are unshared between directions, so the operator instantiates four expansion matrices and two complete attention blocks; a post-norm residual, mean pooling over tokens, concatenation and a projection $\mathbb{R}^{2d} \rightarrow \mathbb{R}^d$ complete it. This accounts exactly for the 584,456 trainable parameters of Proposition 3. The expansion to $K > 1$ is essential by Corollary 1.

All three operators emit exactly $d = 128$ dimensions, so the head that follows has identical capacity (17,544 parameters) in every condition. $\mathbf{z}$ is layer-normalised and passed to Dropout $\rightarrow$ Linear($d \rightarrow d$) $\rightarrow$ GELU $\rightarrow$ Dropout $\rightarrow$ Linear($d \rightarrow 8$). The softmax posterior is retained in full.

D. Language Intelligence Layer

The predicted distribution conditions an instruction-tuned language model, Phi-3-mini-4k-instruct [65]. Rather than passing the top-1 label, the prompt encodes the three highest-probability categories with their posterior mass and instructs the model to hedge when the top two lie within a margin τ (Algorithm 4). The intent is that an ambiguous vocal signal should produce a tentative response rather than a confident misattribution [46], [47]. We do *not* pass the learned representation $\mathbf{z}$ to the language model: a model not trained alongside that network has no basis for interpreting its activations.

We state plainly that this layer is unevaluated here—no human study, no automatic metric, no ablation against top-1-conditioned prompting—and τ is asserted at 0.15 rather than tuned. Algorithm 4 therefore treats τ as a free parameter, to be set from the margin–accuracy curve of Algorithm 6.

E. Application Layer

The pipeline accepts an utterance, returns a predicted category with its posterior, and generates a corresponding response. A demonstration interface is included in the released code. As with Layer 4, no user study supports this layer and we make no claim for it.

Algorithm 3 Unified psychoacoustic–semantic representation**Require:** $\mathbf{x}_p \in \mathbb{R}^{34}$, $\mathbf{x}_h \in \mathbb{R}^{768}$; scalars (S_p, S_h) ; width d ; operator ω **Ensure:** representation $\mathbf{z}$; posterior $\hat{\mathbf{p}}$; gate $\mathbf{g}$

```

1:  $\mathbf{x}_p \leftarrow S_p(\mathbf{x}_p)$ ;  $\mathbf{x}_h \leftarrow S_h(\mathbf{x}_h)$  ▷ fitted on training rows only
2:  $\mathbf{e}_p \leftarrow \text{Enc}_p(\mathbf{x}_p)$ ;  $\mathbf{e}_h \leftarrow \text{Enc}_h(\mathbf{x}_h)$  ▷ both to  $\mathbb{R}^d$ , Eq. (4)
3:  $\mathbf{g} \leftarrow \perp$ 
4: if  $\omega = \text{GATE}$  then
5:    $\mathbf{g} \leftarrow \text{sigm}(\mathbf{W}_g[\mathbf{e}_p; \mathbf{e}_h] + \mathbf{b})$ 
6:    $\mathbf{z} \leftarrow \mathbf{g} \odot \mathbf{e}_p + (\mathbf{1} - \mathbf{g}) \odot \mathbf{e}_h$  ▷ Eq. (14)
7: else if  $\omega = \text{CONCAT}$  then
8:    $\mathbf{z} \leftarrow \mathbf{W}_c[\mathbf{e}_p; \mathbf{e}_h]$ 
9: else
10:   $\mathbf{T}_p \leftarrow \text{reshape}(\mathbf{W}_p \mathbf{e}_p)$ ;  $\mathbf{T}_h \leftarrow \text{reshape}(\mathbf{W}_h \mathbf{e}_h)$ 
11:  assert  $K > 1$  ▷  $K=1$  is gradient-inert, Cor. 1
12:   $\mathbf{A}_{p \rightarrow h}, \mathbf{A}_{h \rightarrow p} \leftarrow \text{Eq. (15)}$ 
13:   $\mathbf{z} \leftarrow \mathbf{W}_o[\mathbf{A}_{p \rightarrow h}; \mathbf{A}_{h \rightarrow p}]$ 
14: end if
15:  $\mathbf{z} \leftarrow \text{LayerNorm}(\mathbf{z})$ ;  $\hat{\mathbf{p}} \leftarrow \text{softmax}(\text{Head}(\mathbf{z}))$ 
16: return  $\mathbf{z}, \hat{\mathbf{p}}, \mathbf{g}$ 

```

Algorithm 4 Posterior-conditioned prompt construction**Require:** posterior $\hat{\mathbf{p}}$; query q ; context h ; margin τ ; model $\mathcal{M}$ **Ensure:** emotion-aware response y

```

1:  $(c_1, c_2, c_3) \leftarrow$  indices of the three largest entries of  $\hat{\mathbf{p}}$ 
2:  $\mu \leftarrow \hat{p}_{c_1} - \hat{p}_{c_2}$  ▷ top-2 margin
3: if  $\mu < \tau$  then
4:    $s \leftarrow$  “evidence is ambiguous between  $c_1$  and  $c_2$ ; acknowledge the uncertainty”
5: else
6:    $s \leftarrow$  “evidence indicates  $c_1$ ; respond to that state”
7: end if
8:  $\pi \leftarrow \text{COMPOSE}(\{(c_i, \hat{p}_{c_i})\}_{i=1}^3, s, h, q)$  ▷ mass stated verbatim
9:  $y \leftarrow \mathcal{M}(\pi)$ ; record  $(\hat{\mathbf{p}}, \mu)$  for calibration
10: return  $y$ 

```

V. RESEARCH METHODOLOGY

A. Methodological Stance

The methodology follows from Section III. Because $\Delta_{\Pi}(\mathcal{A})$ and $\Delta_{\Omega}(\Pi)$ are defined on the same scale, they can be measured in one experimental design by holding one factor fixed while varying the other. We therefore run two orthogonal sweeps on the same code path, the same cached features and the same seeds: a *protocol sweep* with the architecture fixed, and an *architecture sweep* with the protocol fixed at its most defensible setting. A third sweep varies the input streams to address U_p directly. Nothing else changes between runs; the ablations are therefore controlled rather than comparative.

B. Design Decisions and the Alternatives Considered

Table I records each substantive methodological choice, the alternatives available, and the reason for the decision. Two of the decisions work against the apparent strength of our results and are marked accordingly; we record them because a methodology section that lists only favourable choices is not a methodology section.

C. Benchmarking Methodology

Benchmarking in this paper is *protocol-stratified* and *in-harness*, for reasons that follow from (9).

Protocol-stratified external benchmarking. Published figures are grouped by protocol class and never compared across classes (Table XIII). Within a class they remain only loosely comparable, because encoder adaptation, fold count and model-selection practice still differ; we therefore additionally record whether each source reports any measure of dispersion, since a point estimate without dispersion cannot support a ranking at the margins the literature disputes.

In-harness baselines. Any comparison we make on our own behalf is against a system we ran ourselves, on our own folds, in our own code path. Three baselines serve this purpose: chance (12.50%); an MFCC-40 plus random-forest classifier, representing the pre-deep-learning approach; and frozen HuBERT-base alone, which is the ablation of our own psychoacoustic stream and simultaneously the strongest single-encoder reference. The HuBERT-only condition is the one that matters for Objective 1, because it is the term $I(\varphi_h(A); Y)$ in (6) made operational.

What we do not do. We do not report a best-of- n figure, do not select the fusion operator on the test partition, and do not compare our speaker-independent numbers against speaker-dependent published numbers. The last of these would improve our apparent standing considerably.

D. Controlled Decomposition Procedure

Algorithm 5 states the protocol sweep. Three configuration switches vary; everything else is held fixed, so each increment is attributable to a single evaluation-design choice.

E. Assessing the Unified Representation

Aggregate accuracy is a weak instrument for the question of Section III-C: two views can satisfy $U_p > 0$ while their combination moves the mean by less than $\delta_{\min}$. Algorithm 6 therefore measures complementarity on the error sets, and computes the calibration diagnostics Layer 4 requires.

F. Metrics and Statistical Treatment

We report accuracy, macro-averaged F_1 and unweighted average recall (UAR). Because *neutral* has half the support of the other classes, accuracy alone is misleading. For single-split results we report percentile bootstrap 95% confidence intervals over 2000 resamples of the test partition. For cross-validated results we report the mean, the standard deviation across folds (computed with $n - 1$ in the denominator), and

TABLE I

RESEARCH METHODOLOGY: DESIGN DECISIONS, ALTERNATIVES CONSIDERED, AND RATIONALE. ROWS MARKED $\ominus$ ARE CHOICES THAT REDUCE OUR REPORTED PERFORMANCE OR WEAKEN OUR OWN CLAIM, AND WERE ADOPTED ANYWAY.

Decision	Alternatives considered	Adopted	Rationale
Encoder treatment $\ominus$	fine-tuned encoder; weighted layer combination; frozen final layer	frozen final layer, mean-pooled	Isolates the fusion operator and the psychoacoustic stream from encoder adaptation, which is known to dominate [22]. Costs us absolute accuracy; our figures are a floor.
Perceptual descriptors	eGeMAPS-88; ISO 532-1 loudness/sharpness/roughness; cochleagram front-end	34 descriptors (5 F_0 , 5 RMS, 24 Bark)	Small, interpretable and directly tied to critical-band theory. Benchmarked against eGeMAPS on identical folds rather than assumed adequate (Table IX).
Fusion family Ω	attention-only; concatenation-only; late (decision) fusion	all three, at fixed output width	Constraint (4) makes the ablation attributable to the operator. Late fusion was excluded because it changes head capacity.
Primary protocol	random split; single actor-disjoint split; LOSO	actor-disjoint 5-fold	Speaker-disjoint by Definition 1, with dispersion; LOSO would be better (Section VII-D) but was not affordable in the present grid.
Protocol ladder anchor	re-run all rungs per operator	one fixed operator (token cross-attention)	Δ_{Π} is defined at fixed $\mathcal{A}$; by (10) the anchor shifts the result by at most 2.03 points.
Model selection	test partition; validation partition	validation, with test selection reproduced only to measure its optimism	Test selection is the leakage mode most often undisclosed; P2t/P3t exist to quantify it and are never used for a reported result.
Primary metric	accuracy; UAR; macro- F_1	macro- F_1	Fixed in advance because it is also the model-selection criterion, and <i>neutral</i> has half the support.
Multiplicity control $\ominus$	none; Holm across all six tests	Holm within each metric (3 tests)	Accuracy and macro- F_1 on identical folds are strongly dependent; six tests is neither exact nor conservative. This is stricter than no correction.
Claim threshold $\ominus$	report $p < 0.05$; report point estimates	admissibility, Definition 2	Applied to our own results, which fail it (Section VII-D).
Complementarity test	difference of means	error sets, McNemar, per-class Δ_{recall} , gate distribution	$U_p > 0$ can hold below $\delta_{\min}$; means cannot detect it (Section III-C).

the 95% confidence interval of the mean; the last is the inferentially meaningful quantity and the three are labelled distinctly throughout [66].

Paired comparisons between operators use identical folds and report the mean paired difference, its standard deviation, a 95% confidence interval, a paired two-sided t -test, and Holm–Bonferroni adjusted p -values *within* each metric across its three pairwise comparisons [67], [68]. With five folds the exact

Algorithm 5 Controlled decomposition of the protocol effect**Require:** features (X_p, X_h) , labels y , metadata M ; seeds $\mathcal{S}$; fixed architecture $\mathcal{A}$ **Ensure:** accuracy per rung and attributable increments

```

1:  $\mathcal{R} \leftarrow [\text{P1}, \text{P2}, \text{P2t}, \text{P3}, \text{P3t}, \text{P4}]$  ▷ Table II
2: for all rungs  $\rho \in \mathcal{R}$  do
3:   if  $\rho.\text{dup}$  then  $(X, y, M) \leftarrow \text{REPLICATE}(X, y, M; r)$  ▷ restore contamination
4:   for all  $s \in \mathcal{S}$  do
5:      $(I_{\text{tr}}, I_{\text{va}}, I_{\text{te}}) \leftarrow \Pi_\rho(\mathcal{D}, s)$  ▷ Eq. (7)
6:      $I_{\text{sel}} \leftarrow I_{\text{te}}$  if  $\rho.\text{sel} = \text{test}$  else  $I_{\text{va}}$ 
7:      $\theta^* \leftarrow \text{FITFOLD}(\mathcal{A}, I_{\text{tr}}, I_{\text{va}}, I_{\text{sel}}, s)$ 
8:     record accuracy, macro- $F_1$ , UAR on  $I_{\text{te}}$ 
9:   end for
10:   $\mu_\rho \leftarrow$  mean over seeds (and folds for P4)
11: end for
12:  $\Delta_{\text{dup}} \leftarrow \mu_{\text{P1}} - \mu_{\text{P2}}$  ▷ local defect, not a field practice
13:  $\Delta_{\text{spk}} \leftarrow \mu_{\text{P2}} - \mu_{\text{P3}}$  ▷ †
14:  $\Delta_{\text{sel}} \leftarrow \mu_{\text{P3t}} - \mu_{\text{P3}}$  ▷ †
15:  $\Delta_{\text{split}} \leftarrow \mu_{\text{P3}} - \mu_{\text{P4}}$  ▷ †
16: return  $\{\mu_\rho\}, \{\Delta_\bullet\}$ 

```

Algorithm 6 Stream complementarity and posterior calibration**Require:** pooled CV predictions $\{(\hat{\mathbf{p}}_k^{(m)}, y_k)\}$ for $m \in \{p, h, \text{fused}\}$; gates $\{\mathbf{g}_k\}$; bins B **Ensure:** complementarity and calibration report

```

1:  $E_m \leftarrow \{k : \arg \max \hat{\mathbf{p}}_k^{(m)} \neq y_k\}$ 
2:  $J \leftarrow |E_p \cap E_h| / |E_p \cup E_h|$  ▷  $J \rightarrow 1$ : the views fail together
3: report  $|E_h \setminus E_p|, |E_p \setminus E_h|$  ▷ operationalises  $U_p > 0$ , Sec. III-C
4:  $m^* \leftarrow \arg \min_{m \in \{p, h\}} |E_m|$ 
5:  $(b, c) \leftarrow (|E_{m^*} \setminus E_{\text{fused}}|, |E_{\text{fused}} \setminus E_{m^*}|)$ 
6:  $p_{\text{McN}} \leftarrow$  exact binomial test on  $(b, c)$ 
7: report per-class  $\Delta_{\text{recall}}$ , fused  $- m^*$ 
8: report mean and spread of  $\mathbf{g}$  per dimension, per class
9:  $\text{ECE} \leftarrow \sum_b \frac{|B_b|}{N} |\text{acc}(B_b) - \text{conf}(B_b)|$ 
10: plot accuracy against margin  $\mu$ ; set  $\tau^*$  ▷ fixes  $\tau$  in Alg. 4
11: repeat ECE under P1 and P4 ▷ does contamination change calibration?
12: return  $\{J, b, c, p_{\text{McN}}, \text{ECE}_{\text{P1}}, \text{ECE}_{\text{P4}}, \tau^*\}$ 

```

Wilcoxon signed-rank test cannot attain $p < 0.0625$ and is omitted [69]. Design sensitivity follows (11).

VI. EXPERIMENTAL SETUP

A. Evaluation Protocols

Six protocols differ only in partitioning and in which partition selects the model (Table II). *Actor* refers to a RAVDESS performer; *speaker-disjoint* means no actor's utterances appear in more than one partition. Under all protocols, feature standardisation is fitted on the training partition alone.

TABLE II

THE SIX EVALUATION PROTOCOLS. ARCHITECTURE, FEATURES, OPTIMISER, SCHEDULE AND SEEDS ARE IDENTICAL ACROSS ROWS. P2T AND P3T REPRODUCE UNDISCLOSED TEST-SET MODEL SELECTION SO THAT ITS OPTIMISM CAN BE MEASURED; THEY ARE NEVER USED FOR A REPORTED RESULT.

Protocol	Duplicates	Partition	Selection
P1	present (2880)	random 80/20	validation
P2	removed (1440)	random 80/20	validation
P2t	removed	random 80/20	test
P3	removed	actor-disjoint	validation
P3t	removed	actor-disjoint	test
P4	removed	actor-disjoint	validation, 5-fold

TABLE III

PUBLISHED ACTOR ASSIGNMENTS (SEED 42). RAVDESS NUMBERS ACTORS SO THAT ODD IDENTIFIERS ARE MALE AND EVEN ARE FEMALE.

Partition	Actors
P3 train	1, 3, 4, 6, 7, 8, 10, 12, 13, 15, 17, 18, 19, 21, 22, 24
P3 validation	5, 9, 14, 20
P3 test	2, 11, 16, 23
P4 fold 1 (test)	3, 8, 11, 23, 24
P4 fold 2 (test)	6, 7, 14, 18, 19
P4 fold 3 (test)	1, 9, 16, 21, 22
P4 fold 4 (test)	2, 10, 12, 13, 15
P4 fold 5 (test)	4, 5, 17, 20

Algorithm 7 Gender-balanced speaker-disjoint partitioning

Require: metadata M (*actor*, *gender*); fractions τ_{te}, ν ; seed s

Ensure: disjoint I_{tr}, I_{va}, I_{te}

- 1: $A_{tr}, A_{va}, A_{te} \leftarrow \emptyset, \emptyset, \emptyset$
 - 2: **for all** genders g in M **do**
 - 3: $A_g \leftarrow \text{SHUFFLE}(\text{actors of gender } g; s)$
 - 4: $n_{te} \leftarrow \max(1, \lfloor |A_g| \tau_{te} \rfloor)$; $n_{va} \leftarrow \max(1, \lfloor (|A_g| - n_{te}) \nu \rfloor)$
 - 5: append the first n_{te} to A_{te} , the next n_{va} to A_{va} , the rest to A_{tr}
 - 6: **end for**
 - 7: **assert** the three actor sets are pairwise disjoint ▷ overlap fails loudly, not silently
 - 8: $I_{\bullet} \leftarrow \{k : M_k.\text{actor} \in A_{\bullet}\}$
 - 9: **publish** A_{tr}, A_{va}, A_{te} ▷ Table III
 - 10: **return** I_{tr}, I_{va}, I_{te}
-

B. Implementation

Table IV gives the configuration. Optimisation uses AdamW [70] with cosine annealing [71] and label smoothing [72]; cross-entropy is weighted by inverse class frequency to compensate for the half-sized

Algorithm 8 Training one fold without leakage

Require: $X_p, X_h, y; I_{tr}, I_{va}$; selection set I_{sel} (default I_{va}); epochs E ; patience π ; seed s

Ensure: θ^* ; scalars (S_p, S_h)

```

1: SETSEED( $s$ );  $S_m \leftarrow \text{FIT}(X_m[I_{tr}])$  for  $m \in \{p, h\}$  ▷ training rows only
2:  $\mathbf{w} \leftarrow$  inverse class frequency of  $y[I_{tr}]$ 
3:  $b^* \leftarrow -\infty$ ;  $c \leftarrow \pi$ 
4: for  $e = 1$  to  $E$  do
5:   train one epoch (Alg. 3 forward, weighted cross-entropy, gradient clipping)
6:    $m \leftarrow \text{MACROF1}$  on  $I_{sel}$ 
7:   if  $m > b^*$  then
8:      $b^* \leftarrow m$ ;  $\theta^* \leftarrow \theta$ ;  $c \leftarrow \pi$ 
9:   else
10:     $c \leftarrow c - 1$ ; if  $c = 0$  then break
11:   end if
12: end for
13: return  $\theta^*, (S_p, S_h)$  ▷ test partition read exactly once, afterwards

```

TABLE IV
HYPERPARAMETERS AND SIGNAL-PROCESSING CONFIGURATION. HUBERT-BASE IS FROZEN; PARAMETER COUNTS EXCLUDE IT.

Optimiser / LR / weight decay	AdamW / 10^{-3} / 10^{-2}
Schedule / batch / clipping	cosine annealing / 32 / 5.0
Dropout / label smoothing	0.3 / 0.05
Epochs / patience / monitor	100 / 15 / val. macro- F_1
Class weighting	inverse frequency
Width d / tokens K / heads	128 / 4 / 4
Seeds	
Psychoacoustic sample rate	22,050 Hz
HuBERT sample rate	16,000 Hz
FFT / hop / window	2048 / 512 / Hann
F_0 estimator / range	Praat autocorr. / 75–600 Hz
Bark bands / normalisation	24 / unit area
Hardware / software	[GPU, CUDA, versions]

neutral class.

VII. RESULTS

A. Decomposing the Protocol Effect

Table V reports one system under all six protocols. The architecture is held fixed at token cross-attention throughout, that being the configuration in which the ladder was executed; by (10) the choice of anchor changes the measured effect by at most 2.03 points, and re-running the ladder under the adopted gated operator is queued ().

Reading Fig. 2. The vertical axis is test accuracy on a single scale, so the height of each step is the cost of one design decision in the units the field actually reports. The figure should be read from the top down:

TABLE V

EFFECT OF EVALUATION PROTOCOL—ARCHITECTURE (TOKEN CROSS-ATTENTION), FEATURES, OPTIMISER, SCHEDULE AND SEED ARE IDENTICAL ACROSS ROWS; ONLY THE PARTITIONING AND THE SELECTION PARTITION DIFFER. INTERVALS FOR P1 AND P3 ARE PERCENTILE BOOTSTRAP 95% CIs OVER 2000 RESAMPLES OF THE TEST PARTITION; P4 REPORTS MEAN $\pm$ SD ACROSS FOLDS. THE P1 INTERVAL IS GIVEN FOR SYMMETRY ONLY: BECAUSE THAT TEST PARTITION CONTAINS EXACT DUPLICATES OF TRAINING ITEMS, IT DOES NOT ADMIT AN INFERENTIAL INTERPRETATION.

	Protocol	Acc. (%)	Macro- F_1	UAR
P1	duplicated + random	94.79	94.97	95.13
	95% CI	[92.9, 96.5]	[93.1, 96.6]	
P2	deduplicated + random	pending	pending	pending
P2t	+ test selection	pending	pending	pending
P3	actor-disjoint, single	72.50	70.83	73.05
	95% CI	[67.1, 78.3]	[64.8, 76.2]	
P3t	+ test selection	pending	pending	pending
P4	actor-disjoint, 5-fold	64.28	63.42	64.52
	$\pm$ SD	$\pm$ 5.30	$\pm$ 4.45	$\pm$ 4.59
	95% CI of mean	[57.7, 70.9]	[57.9, 68.9]	
	P1 – P4 (total)	30.51	31.56	30.61
	P2 – P4 (attributable)	pending	pending	pending



Fig. 2. Decomposition of the protocol effect. Each rung of Table V is a step, annotated with the increment attributable to the single design choice that separates it from the rung below. The three increments a published system can plausibly incur are solid; Δ_{dup} , which arises from a corrupted local copy rather than a protocol choice, is shown separately. Target file: figures/fig2_decomposition_ladder.pdf.

each descent is what a reader loses by removing one source of optimism, and the total descent is the distance between a number that would pass without comment in most venues and a number a practitioner could rely on. The separation of Δ_{dup} from the remaining three is deliberate—only the lower three are choices the published literature makes.

Three observations follow.

TABLE VI

FUSION-OPERATOR ABLATION UNDER SPEAKER-INDEPENDENT FIVE-FOLD CROSS-VALIDATION (P4), IDENTICAL FOLDS, FEATURES, ENCODER DEPTH, HYPERPARAMETERS AND HEAD CAPACITY. MEAN $\pm$ SD ACROSS FOLDS; 95% CI OF THE MEAN IN BRACKETS. NO VALUE IS BOLDED: THE TOP TWO OPERATORS ARE STATISTICALLY INDISTINGUISHABLE (TABLE VIII).

Fusion	Acc. (%)	Macro-F_1	UAR	Params
Concatenation	66.32 $\pm$ 5.80 [59.1, 73.5]	65.28 $\pm$ 5.30 [58.7, 71.9]	66.31 $\pm$ 5.74 [59.2, 73.4]	187,656
Gated (<i>adopted</i>)	66.25 $\pm$ 4.97 [60.1, 72.4]	65.38 $\pm$ 4.67 [59.6, 71.2]	65.78 $\pm$ 4.64 [60.0, 71.5]	187,656
Cross-attn. ($K=4$)	64.28 $\pm$ 5.30 [57.7, 70.9]	63.42 $\pm$ 4.45 [57.9, 68.9]	64.52 $\pm$ 4.59 [58.8, 70.2]	584,456

First, the contaminated configuration yields 94.79%. Nothing about this figure appears anomalous in isolation: it sits among the 95.83% of Ibrahim *et al.* [57] and the 97.8–99.0% of Jafarzadeh *et al.* [12], and would be reportable without comment. The same system under speaker-disjoint five-fold cross-validation attains 64.28% $\pm$ 5.30—a gap of 30.5 points obtained without changing a line of the model.

We separate that total into two parts, because they carry different weight as criticism of the literature. Δ_{dup} arises from a defect in our own extraction pipeline and not from any practice we can attribute to published work; we report it because it is silent and easy to introduce, not to indict anyone. The remaining three increments—speaker-dependent partitioning, test-set model selection, single-split reporting—are choices that appear in the published record, and their sum $\mu_{P2} - \mu_{P4}$ is the quantity that speaks to the literature.

Second, the single-split speaker-independent figure (72.50%) exceeds the cross-validated mean by 8.2 points and lies outside the latter’s standard deviation. This is a sampling effect, not an inconsistency: with four held-out actors the estimate depends heavily on which four. Per-actor accuracies in that split range from 58.3% to 85.0%, a spread of 26.7 points. We therefore report P4 as our primary result and regard single-split speaker-independent figures as uninterpretable without the fold assignment.

Third, the 95% confidence interval of the P4 mean spans [57.7, 70.9]—a width of 13 points on the quantity the field treats as a single number. That width is itself an argument.

The magnitude is consistent with the leakage literature outside speech [49], [50]. What is specific to acted emotional speech is that the leakage is not accidental: it follows directly from an elicitation design that holds lexical content constant so that emotion is the only varying factor. The property that makes the corpus a clean stimulus set is the property that makes random partitioning catastrophic.

B. Fusion Architecture Ablation

Table VI compares the three operators under P4 on identical folds, and Table VII gives the per-fold values from which every aggregate is computed.

Cross-attention is the weakest of the three on all three metrics. Only one comparison survives multiplicity correction: gated fusion exceeds cross-attention on the primary metric by 1.97 macro- F_1 points

TABLE VII

PER-FOLD TEST ACCURACY (%) UNDER P4. FOLDS ARE IDENTICAL ACROSS OPERATORS (TABLE III); DIFFERENCES ARE THEREFORE PAIRED. FOLD 4, WHOSE HELD-OUT ACTORS ARE 2, 10, 12, 13 AND 15, IS THE HARDEST FOR ALL THREE OPERATORS, WHICH IS WHY THE PAIRED DESIGN IS ESSENTIAL AND A SINGLE SPLIT IS NOT.

Fusion	F1	F2	F3	F4	F5
Concatenation	73.00	69.33	63.33	58.00	67.92
Gated	70.00	70.33	66.67	58.00	66.25
Cross-attention	67.00	69.67	63.67	55.67	65.42

TABLE VIII

PAIRWISE COMPARISONS ACROSS THE FIVE ACTOR-DISJOINT FOLDS OF P4. Δ IS THE MEAN PAIRED DIFFERENCE FROM UNROUNDED FOLD VALUES; $d_z = \Delta/\text{SD}(\Delta)$. HOLM-BONFERRONI IS APPLIED WITHIN EACH METRIC ACROSS ITS THREE COMPARISONS. MACRO- F_1 IS THE PRIMARY METRIC.

Comparison	Metric	Δ	SD	95% CI	p_{Holm}	d_z
Gated – xattn	Macro- F_1	+1.97	0.76	[+1.02, +2.91]	0.013	2.59
Concat – xattn	Macro- F_1	+1.86	2.52	[−1.27, +4.99]	0.348	0.74
Gated – concat	Macro- F_1	+0.10	3.02	[−3.64, +3.85]	0.942	0.03
Gated – xattn	Accuracy	+1.97	1.15	[+0.55, +3.39]	0.055	1.72
Concat – xattn	Accuracy	+2.03	2.61	[−1.21, +5.27]	0.313	0.78
Gated – concat	Accuracy	−0.07	2.44	[−3.10, +2.97]	0.954	−0.03
Gated – xattn	UAR	+1.27	1.02	[+0.00, +2.53]	0.150	1.24
Concat – xattn	UAR	+1.80	2.59	[−1.41, +5.01]	0.390	0.70
Gated – concat	UAR	−0.53	3.39	[−4.74, +3.67]	0.744	−0.16

($p_{\text{Holm}} = 0.013$, $d_z = 2.59$). The corresponding accuracy and UAR comparisons do not survive (0.055, 0.150), and neither concatenation comparison approaches significance. Gated fusion and concatenation are indistinguishable, and the sign of their difference flips between metrics (−0.07 accuracy, +0.10 macro- F_1 , −0.53 UAR)—itself diagnostic that the comparison is noise.

We adopt gated fusion as the reference configuration, and stress that the choice is *not empirically forced*: the two are indistinguishable on every metric ($p_{\text{Holm}} \geq 0.74$), and selecting between them on a tenth of a point would be an instance of the researcher degree of freedom this paper criticises. We prefer the gate for two architectural reasons rather than a performance one. It produces a fixed-width representation independent of the number of streams, so the framework extends without altering downstream capacity; and by Proposition 2 its gate vector is a directly readable per-coordinate allocation between the two views, which concatenation does not expose. A reader who prefers concatenation for its lower cost is equally justified by our results.

Protocol versus architecture. The largest architectural difference is 2.03 accuracy points against a 30.5-point protocol effect measured with the architecture held fixed—a ratio of roughly fifteen to one. Fig. 3 shows both on a common vertical scale.

Reading Fig. 3. The two panels share axis limits; that shared scale is the entire content of the figure and nothing should be read from either panel alone. The left panel spans most of the available range.

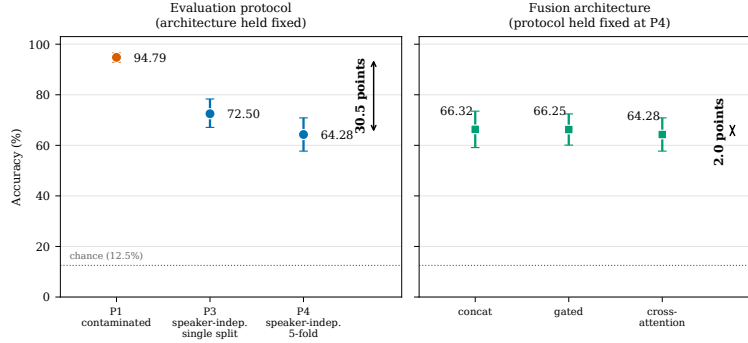


Fig. 3. Evaluation protocol (left) and fusion architecture (right) on a common vertical scale. Varying only the partitioning within one fixed system produces a 30.5-point spread; varying the fusion operator at the most defensible protocol produces 2.0 points. Error bars are 95% confidence intervals (bootstrap over test utterances for P1 and P3; of the mean across folds for P4 and the three operators).

TABLE IX

STREAM ABLATION UNDER P4. IDENTICAL FOLDS, ENCODER DEPTH, LATENT WIDTH $d = 128$ AND HEAD (17,544 PARAMETERS); ONLY THE INPUT STREAM DIFFERS. PARAMETER COUNTS ARE PER ROW BECAUSE ENCODER DEPTH DOES NOT EQUALISE CAPACITY ACROSS INPUT DIMENSIONALITIES ($34 \rightarrow 128$ AGAINST $768 \rightarrow 128$).

Input	Dim	Acc.	Macro- F_1	UAR	Params
Chance	—	12.50	—	12.50	0
Psychoacoustic only	34	pending	pending	pending	39,304
eGeMAPS only	88	pending	pending	pending	pending
HuBERT only	768	pending	pending	pending	133,256
eGeMAPS + HuBERT (gated)	88+768	pending	pending	pending	pending
Both (gated fusion)	34+768	66.25	65.38	65.78	187,656

The right panel collapses to near-flatness, and the error bars of its three operators overlap one another almost completely—so the visual impression that the operators are interchangeable is not an artefact of the scale but the finding itself. A reader who wishes to see the right panel resolved must either accept a scale on which the left panel no longer fits, or collect enough folds to shrink the intervals; Section VII-D quantifies the second option.

C. Stream Ablation and the Value of the Psychoacoustic View

Table IX isolates each stream, and is the direct test of U_p . Unimodal configurations retain identical encoder depth, latent width and head capacity. We additionally report eGeMAPS [4], extracted with openSMILE [25], so the perceptual stream is benchmarked against a standardised parameter set rather than assumed adequate.

The HuBERT-only row is our in-harness frozen-HuBERT baseline (Section V-C), so any comparison we make on our own behalf is against a system we ran on our own folds.

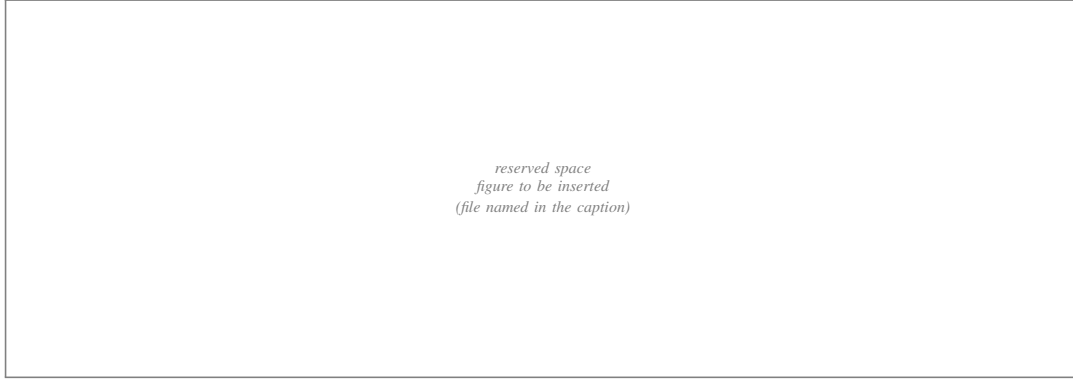


Fig. 4. Complementarity of the two streams under P4, on pooled predictions in which every utterance is classified exactly once by a model that never observed its speaker. *Left*: overlap of the error sets of the two unimodal systems, with the counts each stream alone recovers—the operational form of $U_p > 0$. *Centre*: per-class recall of the fused system minus the stronger unimodal system, so gains offset by losses are visible where aggregate accuracy hides them. *Right*: distribution of the learned gate $\mathbf{g}$ across the 128 dimensions and the eight categories. Target file: `figures/fig5_stream_complementarity.pdf`.

Complementarity beyond the mean. Section VII-D shows this evaluation cannot resolve differences below 3.47 accuracy points, so a psychoacoustic contribution smaller than that is invisible in Table IX whether or not $U_p > 0$. Algorithm 6 therefore examines the error sets directly. Fig. 4 reports the result.

Reading Fig. 4. The left panel is the panel that answers Objective 1, and the quantity to look at is the two crescents rather than the intersection: a large intersection with empty crescents means the views fail on the same utterances and the perceptual stream is redundant, while populated crescents mean $U_p > 0$ regardless of what the accuracy column says. The centre panel guards against a specific misreading of a null aggregate result—a stream that raises recall on two categories and lowers it on two others produces zero net accuracy while changing the model substantially. The right panel is diagnostic rather than evaluative: a gate concentrated near zero means the model has learned to ignore the perceptual stream, which would corroborate redundancy from the model’s own parameters rather than from its outputs.

D. What This Evaluation Can Resolve

Applying (11) with the observed standard deviation of paired fold-wise accuracy differences, $\overline{\text{SD}(\Delta)} = 2.07$ points averaged over the three accuracy comparisons of Table VIII, a two-sided paired t -test at $\alpha = 0.05$ attains 80% power only for $|d_z| \geq 1.68$, giving $\delta_{\min}(5) = \mathbf{3.47}$ accuracy points. The same calculation gives 2.06 points at ten folds and 1.23 points under leave-one-speaker-out with 24 folds (Fig. 5).

Reading Fig. 5. The curve falls as $1/\sqrt{n}$ modulated by the t critical value, so the steep early portion is where additional folds buy the most resolution: moving from five to ten folds recovers 1.4 points of sensitivity, while moving from ten to twenty-four recovers only 0.8. The two horizontal reference lines are our own measured effects. The figure should be read as a design tool rather than a result—given a difference one hopes to detect, it returns the number of folds required, and given a number of folds it returns the smallest claim that would be admissible under Definition 2.

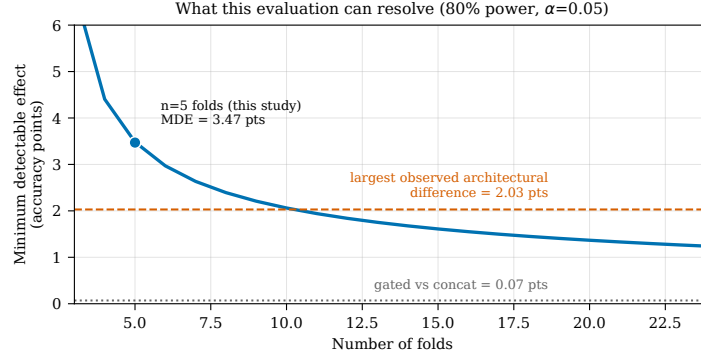


Fig. 5. Design sensitivity as a function of fold count, at 80% power and $\alpha = 0.05$, using the observed standard deviation of paired fold-wise accuracy differences. At five folds the design resolves 3.47 accuracy points. The largest difference between any two fusion operators (2.03 points, dashed) and the difference between the top two (0.07 points, dotted) both fall below it.

By Definition 2, the difference between the best and worst fusion operators (2.03 points) and between the top two (0.07 points) are both **inadmissible** under our own design. This is the operational content of the paper’s central claim: not merely that architectural effects are smaller than protocol effects, but that on this corpus, under a protocol a careful author would defend, they are smaller than the evaluation can measure.

One apparent tension should be addressed, since it is easily misread. Table VIII reports a 1.97-point macro- F_1 difference as significant after correction, below the 3.47-point threshold. There is no contradiction: the threshold uses the *average* paired standard deviation across comparisons (2.07 points), whereas that comparison exhibited an unusually small one (0.76 points). Design sensitivity characterises what an evaluation can be relied upon to detect in general, not what it detected in one favourable case—and with $n = 5$ an SD estimate of 0.76 is itself imprecise. We report the significant comparison and decline to build on it.

E. Per-Class Performance and Confusions

Table X contrasts per-class results under the contaminated and speaker-independent protocols for the same system.

Happy is the dominant failure mode under P3, with recall 0.375; its errors distribute into *angry* (8 of 32), *surprised* (6) and *neutral* (5). *Sad* has the second-lowest recall (0.500). *Neutral* shows high recall but low precision (0.448), indicating that the class-weighted objective over-predicts the minority class. Section VIII-D accounts for both. Class supports are small and the intervals correspondingly wide: recall of 0.375 on 32 utterances carries a binomial 95% interval of roughly [0.21, 0.57].

The ΔF_1 column carries a finding of its own. Contamination does not inflate the classes uniformly: it flatters *happy* by 0.404 and *neutral* by 0.396 while flattering *fearful* by only 0.083. The two classes a speaker-independent system most needs to fix are precisely the two a contaminated evaluation reports as solved.

TABLE X

PER-CLASS F_1 UNDER THE CONTAMINATED PROTOCOL P1 ($n = 576$) AND THE SPEAKER-INDEPENDENT SINGLE SPLIT P3 ($n = 240$, ACTORS 2, 11, 16, 23), SAME ARCHITECTURE. Δ IS THE AMOUNT BY WHICH CONTAMINATION FLATTERS EACH CLASS. THE FAILURE THAT DOMINATES SPEAKER-INDEPENDENT PERFORMANCE, *happy*, IS THE CLASS CONTAMINATION CONCEALS MOST.

Emotion	P1 (contaminated)		P3 (speaker-disjoint)			ΔF_1
	Rec.	F_1	Prec.	Rec.	F_1	
Angry	1.000	0.950	0.692	0.844	0.761	+0.189
Calm	0.961	0.980	0.846	0.688	0.759	+0.221
Disgust	0.935	0.966	0.824	0.875	0.848	+0.118
Fearful	0.922	0.922	0.867	0.812	0.839	+0.083
Happy	0.909	0.915	0.800	0.375	0.511	+0.404
Neutral	1.000	0.974	0.448	0.812	0.578	+0.396
Sad	0.948	0.930	0.696	0.500	0.582	+0.348
Surprised	0.935	0.960	0.682	0.938	0.789	+0.171
Macro	0.951	0.950	0.732	0.730	0.708	+0.242

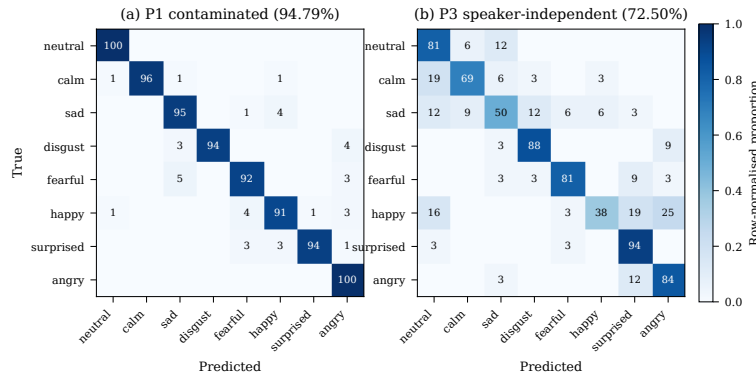


Fig. 6. Row-normalised confusion matrices under (a) the contaminated random-split protocol P1 and (b) the speaker-independent protocol P3. Classes are ordered by arousal and valence rather than alphabetically.

Reading Fig. 6. The two panels are the same system under two protocols, so every difference between them is caused by the evaluation design and none by the model. Panel (a) is what near-perfect diagonal structure looks like when a classifier can identify the talker rather than the emotion, and it is worth dwelling on how unremarkable it appears. Panel (b) is informative in its off-diagonal structure: because the classes are ordered by arousal and valence, the errors form a visible block among the high-arousal states of opposing valence—*happy*, *angry* and *surprised*—which is the acoustically predicted confusion and is exactly what Section VIII-D attributes to mean-pooling and to the absence of voice-quality descriptors. The block is absent from panel (a) not because the model has solved it but because it never had to.

TABLE XI

CONTAMINATION CONCEALS THE FAIRNESS FAILURE. SAME ARCHITECTURE, SAME FEATURES, SAME SEED; ONLY THE EVALUATION PROTOCOL DIFFERS. A DISPARITY AUDIT RUN UNDER P1 WOULD REPORT NEAR-PARITY FOR A SYSTEM THAT IS 21.6 POINTS WORSE ON MALE SPEAKERS.

	P1 (contaminated)	P3 (speaker-disjoint)	Ratio
Female accuracy	96.0	83.3	—
Male accuracy	93.5	61.7	—
Gender gap	2.5	21.6	8.6×
Per-actor min	85.7	58.3	—
Per-actor max	100.0	85.0	—
Actor spread	14.3	26.7	1.9×
Worst-class F_1	0.915 (<i>happy</i>)	0.511 (<i>happy</i>)	1.8×

F. Contamination Conceals Speaker and Gender Disparity

Under P3, per-actor accuracy ranges from 58.3% (actor 23) to 85.0% (actor 2), and accuracy is 83.3% on female speakers against 61.7% on male—a 21.6-point disparity, comparable to asymmetries reported in fairness analyses of transformer-based SER [19].

The same system evaluated under P1 tells a different story (Table XI). The gender gap shrinks to 2.5 points and the per-actor spread narrows from 26.7 to 14.3 points across all 24 actors. A fairness audit conducted on the contaminated protocol would conclude the system is close to parity. It is not.

This extends the paper’s argument beyond reported accuracy. A protocol that inflates a headline number by 30 points also suppresses the two audits—per-group disparity and per-class failure—on which responsible deployment depends, and suppresses them by a larger relative factor than it inflates the headline. We report the disparity rather than correcting for it; per-speaker normalisation is an obvious mitigation. Two cautions apply: the P3 figure rests on four held-out actors and should be recomputed across all 24 from pooled P4 predictions (), and the magnitude of the variation is itself a further argument for cross-validated reporting.

G. Empirical Confirmation of the Degeneracy Result

Proposition 1 and Corollary 1 predict that the framework as originally specified—cross-attention applied directly to pooled stream vectors, so that each sequence has length one—is exactly gradient-inert. We confirmed this on the original implementation. Scaling the psychoacoustic input by a factor of 100 while holding the HuBERT input fixed changed the output logits by 0.0 to machine precision, and the accumulated gradient with respect to every parameter of the psychoacoustic encoder was exactly zero.

Two consequences deserve emphasis. First, any attention map derived from such a configuration is uninformative: every weight is 1.0 by construction, and interpreting it as evidence of learned cross-stream interaction is unwarranted—a specific instance of the concern raised by Bibal *et al.* [37]. Second, the failure is silent. The model trains, the loss decreases, accuracy is competitive, and no diagnostic short

TABLE XII

COMPUTATIONAL PROFILE OF THE TRAINABLE STACK, EXCLUDING THE FROZEN HUBERT ENCODER. MACs ARE MULTIPLY-ACCUMULATE OPERATIONS PER UTTERANCE, COUNTING LINEAR PROJECTIONS AND ATTENTION PRODUCTS. THE TRAINABLE STACK IS NEGLIGIBLE BESIDE ONE FROZEN-ENCODER FORWARD PASS (≈ 16 GMAC FOR A 3.7 S UTTERANCE), SO OPERATOR CHOICE IS A MEMORY AND PARAMETER QUESTION RATHER THAN A THROUGHPUT ONE.

Configuration	Fusion params	Total params	MACs	Rel.
Psychoacoustic only	—	39,304	38,144	$0.21\times$
HuBERT only	—	133,256	132,096	$0.71\times$
Concatenation	32,896	187,656	185,600	$1.00\times$
Gated (<i>adopted</i>)	32,896	187,656	185,600	$1.00\times$
Cross-attn. ($K=4$)	429,696	584,456	980,224	$5.28\times$

of a gradient check reveals that half the architecture is inert. The assertion at line 11 of Algorithm 3 exists to make it loud.

Correcting the mechanism by expanding each stream into $K = 4$ tokens restores gradient flow to both encoders. It does not make cross-attention competitive: Table VI shows the corrected mechanism underperforming a single sigmoid gate at $3.1\times$ the parameters and $5.3\times$ the multiply-accumulate cost. We have not swept K ; the claim should be read as holding at $K = 4$ rather than for all K .

H. Computational Complexity

Table XII instantiates Proposition 3. Counts are exact, derived analytically and confirmed against the implementation.

Three points follow. The gate costs exactly what concatenation costs, so Proposition 2’s interpretability advantage is free. Token cross-attention costs $3.11\times$ the parameters and $5.28\times$ the operations for the worst accuracy of the three, which is the strongest practical statement the paper can make about it. And because the frozen encoder dominates by four to five orders of magnitude, none of these differences is a deployment consideration—the case for the gate rests on parameter economy and interpretability, not on latency, and we decline to argue otherwise.

I. Performance Matrix and Benchmarking

Table XIII consolidates every system we ran and every published figure we compare against, under the protocol-stratified discipline of Section V-C.

Three features of Table XIII carry the benchmarking argument.

First, the highest score in the table belongs to a random-split system, and the fifth-highest to MFCC features with a random forest—a pre-deep-learning baseline that outscores every speaker-independent entry, including systems using encoders that did not exist when it was designed. Under protocol-blind comparison one would conclude that SER has regressed for a decade. This is the clearest available demonstration that cross-protocol tabulation is not merely imprecise but inverts the ordering.

TABLE XIII

PERFORMANCE MATRIX. FIGURES ARE **NOT** COMPARABLE ACROSS PROTOCOL CLASSES, AND WITHIN A CLASS REMAIN ONLY LOOSELY COMPARABLE BECAUSE ENCODER ADAPTATION, FOLD COUNT AND MODEL-SELECTION PRACTICE DIFFER. *Disp.* RECORDS WHETHER THE SOURCE REPORTS ANY MEASURE OF VARIABILITY. ROWS MARKED *in-harness* WERE RUN BY US, ON THE FOLDS OF TABLE III, IN THE RELEASED CODE. LSO DENOTES LEAVE-SPEAKER-OUT; LOSO LEAVE-ONE-SPEAKER-OUT.

System	Protocol	Acc. (%)	Macro- F_1	UAR	Disp.	Params	Source
<i>(a) Speaker-dependent — random utterance partition. Not comparable with (b).</i>							
wav2vec2 + classifier	random 80/20	99.0	—	—	✓	—	[12]
HuBERT + classifier	random 80/20	97.8	—	—	✓	—	[12]
Hybrid fusion (A+V)	random split	95.83	—	—	×	—	[57]
PAER-LLM, cross-attn. (P1)	duplicated + random	94.79	94.97	95.13	✓	584,456	<i>in-harness</i>
MFCC-40 + random forest	random split	92.36	—	—	×	—	<i>in-harness</i> ^a
emotion2vec	random 10-fold	82.43	—	—	×	—	[21]
<i>(b) Speaker-independent. Our primary result is PAER-LLM (P4).</i>							
wav2vec2 (weighted layers)	fixed actor split	—	—	84.3	✓	—	[15]
CAMuLeNet	10-fold LSO	82.3	—	—	×	—	[11]
XLSR-wav2vec2 + MLP	subj.-wise 5-fold	81.82	—	—	✓	—	[58]
Whisper-Medium	10-fold LSO	71.3	—	—	×	—	[11]
PAER-LLM, single split (P3)	actor-disjoint, 1 split	72.50	70.83	73.05	✓	584,456	<i>in-harness</i>
PAER-LLM, concat (P4)	actor-disj. 5-fold	66.32	65.28	66.31	✓	187,656	<i>in-harness</i>
PAER-LLM, gated (P4)	actor-disj. 5-fold	66.25	65.38	65.78	✓	187,656	<i>in-harness</i>
WavLM	10-fold LSO	65.4	—	—	×	—	[11]
PAER-LLM, cross-attn. (P4)	actor-disj. 5-fold	64.28	63.42	64.52	✓	584,456	<i>in-harness</i>
HuBERT	10-fold LSO	58.7	—	—	×	—	[11]
HuBERT only (ours)	actor-disj. 5-fold	pending	pending	pending	✓	133,256	<i>in-harness</i>
Psychoacoustic only (ours)	actor-disj. 5-fold	pending	pending	pending	✓	39,304	<i>in-harness</i>
Hybrid fusion (A+V)	LOSO	48.06	—	—	✓	—	[57]
wav2vec2.0	10-fold LSO	39.6	—	—	×	—	[11]
Chance	—	12.50	—	12.50	—	0	—

^aOur own pre-deep-learning baseline. Its protocol is a random split; like the published rows of class (a) it is not comparable with class (b), and we include it to show that a 1990s feature set under a contaminated protocol outscores every speaker-independent system in the table.

Second, class (b) spans 44.7 points—wider than class (a)’s 16.6—so protocol annotation alone does not make figures comparable. Fold count, encoder adaptation and selection practice still differ, and the two entries in class (b) that exceed our result by the largest margin [11], [58] both adapt the encoder, which we deliberately do not.

Third, the *Disp.* column is mostly empty. A point estimate without dispersion cannot support a ranking at the one-to-three-point margins the literature disputes, which is the quantitative form of the recommendation in Section VIII-B.

VIII. DISCUSSION

A. Overall Discussion

Read together, the results describe a field measuring below its own noise floor and not reporting that it is doing so.

The protocol effect is 30.5 accuracy points; the architecture effect at the most defensible protocol is 2.03; the design’s minimum detectable effect is 3.47. These three numbers are the paper in miniature. The first exceeds the second by a factor of fifteen, which means that any comparison across papers with different partitioning schemes is dominated by a nuisance factor. The third exceeds the second, which means that even a comparison *within* one careful harness cannot resolve differences of the size the literature routinely reports as contributions. Our own framework’s contribution is subject to the same bound, and we have applied it to ourselves in Section VII-D rather than exempting the proposed method.

Three further findings sharpen this. The degeneracy result (Proposition 1) shows that a widely used fusion construction can be not merely weak but formally inert, undetectably so without a gradient check, which means some published dual-stream results may describe single-stream models. The convex-hull result (Proposition 2) shows that the two operators that perform identically are not the same function class, so their empirical tie is a statement about the data rather than about the operators. And the fairness result (Section VII-F) shows that contamination suppresses disparity audits by a larger relative factor ($8.6\times$) than it inflates headline accuracy ($1.5\times$), so a protocol chosen for convenience propagates into claims about who a system works for.

What the results do *not* show is that psychoacoustic descriptors are redundant. Section III-C distinguishes $U_p = 0$ from $U_p > 0$ -but-unmeasurable, and this evaluation cannot separate them; that is precisely why Algorithm 6 measures error sets rather than means. The honest position on Objective 1 is that the framework is specified, its operators are characterised analytically, its cost is quantified, and the representational question it was built to answer is currently bounded by the measurement apparatus rather than by the representation.

B. Implications for the Literature

If a 30-point swing is obtainable by partitioning alone, a reported accuracy without a stated protocol carries almost no information. We do not suggest speaker-dependent evaluation is never appropriate—for a system intended to adapt to a known enrolled user it may be the relevant condition—but the condition must be stated, and results obtained under it must not be compared with speaker-independent figures. The confusion is compounded by ambiguous terminology: “leave-one-out” and “ k -fold cross-validation” are used for both utterance-level and speaker-level partitioning.

We recommend that publications on acted emotional speech corpora report, at minimum: (i) the number of unique recordings alongside the number of rows used; (ii) whether partitioning is speaker-disjoint, with the actor assignment published; (iii) dispersion across folds, not a point estimate; (iv) which partition was used for model selection; and (v) the design sensitivity of the evaluation, so readers can judge whether a reported improvement is measurable. Items (ii) and (iii) were the norm established in 2009 [54]. Adopting

an externally defined split such as EmoBox [56] satisfies (i)–(ii) without requiring each author to invent a protocol.

C. Why the Fusion Behaves as It Does

Proposition 2 establishes that the gate and concatenation span distinct function classes; their empirical tie therefore requires explanation from the data. Attention is a mechanism for *selecting among* elements of a sequence conditioned on a query, and its advantage arises when there is structure to select over. After mean-pooling, each stream is a single fixed-length vector with no temporal correspondence to the other, so there is nothing to align. The K -token expansion manufactures a sequence by linear projection, but the tokens are deterministic functions of the same pooled vector and carry no independent information; attention over them can only recombine what a linear layer already provides, at $5.28\times$ the cost. On this reading the gate and the concatenation are not merely adequate but sufficient: with two vectors and no correspondence structure, the useful combination operators are essentially per-coordinate reweightings, and both span that space from opposite sides of Proposition 2.

This predicts that cross-attention should become advantageous as soon as genuine sequences are retained—frame-level HuBERT states aligned against frame-level psychoacoustic contours—consistent with gains reported for temporally-resolved attentive pooling [73]. It also identifies where the unified representation should be built next: not in a better operator over pooled vectors, but at the frame level, where the two views have something to align.

D. Why Particular Emotions Fail

Happy fails for over-determined reasons. Acoustically, happiness in acted speech is a high-arousal state whose principal correlates—elevated mean F_0 , increased F_0 range, raised intensity, faster rate—are shared with anger and surprise; the cues that distinguish them are spectral-balance and voice-quality measures (spectral tilt, harmonics-to-noise ratio, jitter, shimmer) and temporal contour shape, none of which survive mean-pooling and none of which appear in our 34-descriptor set. Representationally, mean-pooling discards the rise–fall contour that distinguishes surprise from sustained joy. At the corpus level, RAVDESS’s *happy* is a portrayed rather than felt state. We predict the failure would be substantially reduced by retaining temporal structure and adding voice-quality descriptors—a prediction the eGeMAPS row of Table IX partially tests, since eGeMAPS includes exactly those measures.

Neutral is over-predicted as a direct consequence of the inverse-frequency class weighting adopted to compensate for its halved support. The alternatives—no weighting, resampling, or a margin-based loss—trade neutral recall against neutral precision. It is also worth noting that *neutral* is the acoustic origin of the arousal–valence space, so any weakly realised emotional colouring falls near it; some part of the low precision is substantive rather than an artefact of weighting.



Fig. 7. Posterior calibration under the contaminated and speaker-independent protocols. *Left*: reliability diagrams for P1 and P4 with expected calibration error, on pooled predictions. *Right*: accuracy conditioned on the top-1 minus top-2 posterior margin, from which the hedging threshold τ of Algorithm 4 is set rather than asserted. Target file: figures/fig7_reliability.pdf.

E. The Language-Intelligence Layer

This layer is the least evidenced part of the work, and we treat it accordingly. Conditioning a language model on a posterior rather than a label is a defensible design, but as implemented it is posterior-formatted prompting with an asserted threshold, and we make no claim for it.

The experiment that would make it load-bearing is the calibration analysis already specified in Algorithm 6. Computing expected calibration error and a reliability diagram on pooled cross-validated predictions would establish whether the posterior carries information about correctness at all; conditioning accuracy on the top-1 minus top-2 margin would allow τ to be set rather than asserted. Computing the same quantities under P1 and P4 would test a stronger hypothesis: that a contaminated protocol produces a model that is not merely more accurate but differently *calibrated*, and therefore one whose downstream interaction policy is tuned to a confidence it does not possess. Given that contamination already distorts the fairness audit by $8.6\times$ (Section VII-F), we regard this as likely. Fig. 7 reserves the space.

Reading Fig. 7. In the left panel the diagonal is perfect calibration; a curve below it marks overconfidence. The comparison to make is not between either curve and the diagonal but between the two curves: if P1 and P4 are displaced differently, then a hedging policy tuned on a contaminated evaluation will hedge at the wrong confidence in deployment, independently of how much accuracy the contamination added. The right panel is the design curve for τ : the value at which accuracy rises steeply is the margin below which the system should decline to assert a state, and reading it off the plot replaces the asserted 0.15 with a measured quantity.

What this layer does not do is learn multimodal grounding. Approaches that align speech representations into the language model’s embedding space [43], [44] achieve something categorically different, and we regard that as the appropriate direction for this component.

F. Practical Implications

For practitioners, published RAVDESS accuracies are not a guide to deployed performance on unheard speakers, and the gap is not a matter of a few points. A system selected on a 97% benchmark figure may deliver in the mid-sixties on a new talker, and—by Section VII-F—may deliver considerably worse than that for one gender. For affective interfaces in mental-health triage, driver-state monitoring or call-centre analytics, the procurement question is not “what accuracy does it report” but “under what partition, over how many speakers, with what dispersion, and with what per-group breakdown”. For system builders, the results argue for spending effort on encoder adaptation and temporal modelling, where the literature shows double-digit effects, rather than on fusion-operator design over pooled vectors, where we measure effects below the noise floor and prove one popular operator inert.

G. What Generalises and What Does Not

The numerical results are RAVDESS-specific. The analytical results of Section III-F are corpus-independent: the degeneracy of single-token attention is a property of the softmax, the convex-hull confinement a property of the sigmoid, and the cost ordering a property of the parameterisation. The *magnitude* of the protocol effect is corpus-dependent and we expect it near an upper bound here, since RAVDESS’s factorial design maximises the near-duplicate structure available to a random split; on corpora with lexical variety and more speakers such as CREMA-D [74] the effect should be smaller, and on smaller corpora such as EmoDB [75] larger [50]. The *ordering*—that protocol effects exceed architectural effects—we expect to be robust, because architectural effects in this literature are consistently reported at the 1–3 point scale while speaker-dependence effects are consistently reported at 20–40 points wherever both protocols are run on one system [57], [59].

H. Future Work

Multi-corpus replication of the decomposition on CREMA-D [74], EmoDB [75] and IEMOCAP [76] using EmoBox splits [56]; frame-level rather than pooled fusion with genuine cross-attention, which Section VIII-C identifies as the setting where the unified representation has room to improve; middle-layer and weighted-layer representations [17], [18]; standardised psychoacoustic metrics (ISO 532-1 loudness, sharpness, roughness [61]) in place of Bark energies; per-speaker normalisation to address the gender disparity; deep LLM integration via encoder alignment [43], [44] with human evaluation; and extension of the decomposition to naturalistic corpora following the Interspeech 2025 challenge protocol [55].

IX. LIMITATIONS

- 1) **Single corpus.** All results are on RAVDESS. Acted speech from 24 actors reading two fixed sentences is not representative of spontaneous emotional expression, and cross-corpus generalisation is considerably weaker [59].
- 2) **Seed variance unmeasured in the reported tables.** The results derive from a single seed. Initialisation variance is plausibly comparable to the architectural differences we report [52], and the

multi-seed grid () is required before Table VIII can be treated as settled. If seed variance swamps the fusion differences, that strengthens rather than weakens the central claim.

- 3) **Five folds.** With $n = 5$ the exact Wilcoxon test cannot reach conventional significance and $\delta_{\min} = 3.47$ points. Leave-one-speaker-out would reduce this to 1.23 and is standard on this corpus.
- 4) **The representational claim is bounded by the evaluation.** The contribution of the psychoacoustic stream can be established from means only if it exceeds $\delta_{\min}$; below that, Algorithm 6 rather than Table IX carries the argument.
- 5) **Protocol ladder anchored on one operator.** Table V is executed under token cross-attention. By (10) the anchor shifts the result by at most 2.03 points, but symmetry with Table VI would be preferable.
- 6) **Frozen encoder, final-layer pooling.** We use mean-pooled HuBERT-base without fine-tuning, discarding temporal structure and forgoing the largest available source of improvement [22], and we pool the final hidden state though middle layers carry more paralinguistic information [17], [18]. Our absolute figures are a floor.
- 7) **No K sweep.** The conclusion that cross-attention is uncompetitive is established at $K = 4$ only.
- 8) **Bark band placement.** Our 24 bands are equal-width over $[0, z(f_{\text{Nyq}})]$ rather than the canonical Zwicker bands, and the 22.05 kHz analysis rate discards the upper 1.14 Bark of the audible range.
- 9) **Gender disparity reported but not addressed,** and computed on four held-out actors (Section VII-F).
- 10) **Latency and memory unprofiled.** Table XII reports parameters and MACs analytically, not wall-clock or peak memory.
- 11) **Language layer unevaluated.** No human study, automatic metric or ablation supports Layer 4, and τ is asserted rather than tuned.

Several of these—single corpus, single seed, five folds, absent dispersion on some quantities—are precisely the reporting weaknesses this paper criticises. We state them explicitly rather than omitting them, which is the standard we advocate.

X. CONCLUSION

We proposed PAER-LLM, a five-layer framework that unifies perceptually motivated psychoacoustic descriptors with frozen self-supervised speech representations through a learned per-dimension gate, and conditions an instruction-tuned language model on the resulting posterior rather than on a discarded top-1 label. We formalised the unified-representation objective, the information-theoretic condition under which the perceptual view can contribute, the evaluation protocol as an operator, and an admissibility criterion that a representational claim must satisfy.

Evaluating the framework under six protocols differing only in partitioning and model selection, we measured a 30.5-point spread in reported accuracy with the architecture held fixed. Under the same conditions the largest difference between three fusion architectures was 2.03 points—below the 3.47-point minimum detectable effect of the evaluation, and therefore inadmissible under our own criterion.

Contamination additionally conceals a 21.6-point gender disparity, reporting it as 2.5, and conceals the dominant per-class failure entirely.

We proved that single-token cross-modal attention is analytically degenerate: it silently reduces a dual-stream model to a single-stream one, and no diagnostic short of a gradient check reveals it. We further showed that gated fusion and concatenation span distinct function classes despite performing identically, and that the corrected multi-token attention costs $3.1\times$ the parameters and $5.3\times$ the operations for the worst accuracy of the three.

Our speaker-independent result, $66.25\% \pm 4.97$ accuracy with frozen HuBERT-base features and gated fusion, is reported with dispersion, confidence intervals, per-speaker and per-gender breakdowns and published fold assignments. We release the evaluation harness, including the contaminated-protocol reproduction.

The field does not lack architectural proposals. It lacks the evaluation discipline required to tell which of them work.

REPRODUCIBILITY STATEMENT

RAVDESS is publicly available under CC BY-NC-SA 4.0 [10] from <https://doi.org/10.5281/zenodo.1188976>. The complete implementation—feature extraction, the three fusion operators, the actor-disjoint splitting utilities, the contaminated-protocol reproduction, the evaluation harness and all configuration files—is available at [repository URL] (archived at Zenodo, DOI [DOI]). The repository contains the actor-to-fold assignment of Table III as a machine-readable manifest, the per-fold prediction files from which every table and figure is generated, the scripts that regenerate them, and the corpus-integrity utility of Algorithm 1. The environment is pinned in `requirements.txt`.

The numbers are cross-consistent in ways checkable without the code: in Table X, the support-weighted mean recall of the P3 column equals the P3 accuracy of Table V (72.50%), its macro-averaged recall equals the P3 UAR (73.05%), and its macro-averaged F_1 equals the P3 macro- F_1 (70.83%), each to two decimal places; and the per-fold values of Table VII reproduce every mean, standard deviation and paired difference in Tables VI and VIII.

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[acknowledgements, or delete this section]

REFERENCES

- [1] T. M. Wani, T. S. Gunawan, S. A. A. Qadri, M. Kartiwi, and E. Ambikairajah, “A comprehensive review of speech emotion recognition systems,” *IEEE Access*, vol. 9, pp. 47 795–47 814, 2021.
- [2] S. Latif, R. Rana, S. Khalifa, R. Jurdak, J. Qadir, and B. W. Schuller, “Survey of deep representation learning for speech emotion recognition,” *IEEE Transactions on Affective Computing*, vol. 14, no. 2, pp. 1634–1654, 2023.
- [3] S. K. Khare, V. Blanes-Vidal, E. S. Nadimi, and U. R. Acharya, “Emotion recognition and artificial intelligence: A systematic review (2014–2023) and research recommendations,” *Information Fusion*, vol. 102, p. 102019, 2024.

- [4] F. Eyben, K. R. Scherer, B. W. Schuller, J. Sundberg, E. André, C. Busso, L. Y. Devillers, J. Epps, P. Laukka, S. S. Narayanan, and K. P. Truong, "The Geneva minimalistic acoustic parameter set (GeMAPS) for voice research and affective computing," *IEEE Transactions on Affective Computing*, vol. 7, no. 2, pp. 190–202, 2016.
- [5] E. Zwicker, "Subdivision of the audible frequency range into critical bands (Frequenzgruppen)," *The Journal of the Acoustical Society of America*, vol. 33, no. 2, p. 248, 1961.
- [6] A. Baevski, Y. Zhou, A. Mohamed, and M. Auli, "wav2vec 2.0: A framework for self-supervised learning of speech representations," in *Advances in Neural Information Processing Systems 33 (NeurIPS)*, 2020.
- [7] W.-N. Hsu, B. Bolte, Y.-H. H. Tsai, K. Lakhotia, R. Salakhutdinov, and A. Mohamed, "HuBERT: Self-supervised speech representation learning by masked prediction of hidden units," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 29, pp. 3451–3460, 2021.
- [8] S. Chen, C. Wang, Z. Chen, Y. Wu, S. Liu, Z. Chen, J. Li, N. Kanda, T. Yoshioka, X. Xiao, J. Wu, L. Zhou, S. Ren, Y. Qian, Y. Qian, M. Zeng, X. Yu, and F. Wei, "WavLM: Large-scale self-supervised pre-training for full stack speech processing," *IEEE Journal of Selected Topics in Signal Processing*, vol. 16, no. 6, pp. 1505–1518, 2022.
- [9] S. Dixit, F. Catania, D. M. Low, G. Elbanna, and S. S. Ghosh, "Explaining deep learning embeddings for speech emotion recognition by predicting interpretable acoustic features," 2024.
- [10] S. R. Livingstone and F. A. Russo, "The Ryerson audio-visual database of emotional speech and song (RAVDESS): A dynamic, multimodal set of facial and vocal expressions in North American English," *PLOS ONE*, vol. 13, no. 5, p. e0196391, 2018.
- [11] A. Goel, M. Hira, and A. Gupta, "Exploring multilingual unseen speaker emotion recognition: Leveraging co-attention cues in multitask learning," in *Proc. Interspeech*, 2024, pp. 2340–2344.
- [12] P. Jafarzadeh, A. M. Rostami, and P. Choobdar, "Speaker emotion recognition: Leveraging self-supervised models for feature extraction using wav2vec2 and HuBERT," 2024.
- [13] C. Nießl, M. Herrmann, C. Wiedemann, G. Casalicchio, and A.-L. Boulesteix, "Over-optimism in benchmark studies and the multiplicity of design and analysis options when interpreting their results," *WIREs Data Mining and Knowledge Discovery*, vol. 12, no. 2, 2022.
- [14] S.-w. Yang, P.-H. Chi, Y.-S. Chuang, C.-I. J. Lai, K. Lakhotia, Y. Y. Lin, A. T. Liu, J. Shi, X. Chang, G.-T. Lin, T.-H. Huang, W.-C. Tseng, K.-t. Lee, D.-R. Liu, Z. Huang, S. Dong, S.-W. Li, S. Watanabe, A. Mohamed, and H.-y. Lee, "SUPERB: Speech processing universal PERFORMANCE benchmark," in *Proc. Interspeech*, 2021, pp. 1194–1198.
- [15] L. Pepino, P. Riera, and L. Ferrer, "Emotion recognition from speech using wav2vec 2.0 embeddings," in *Proc. Interspeech*, 2021, pp. 3400–3404.
- [16] A. Pasad, J.-C. Chou, and K. Livescu, "Layer-wise analysis of a self-supervised speech representation model," in *Proc. IEEE Automatic Speech Recognition and Understanding Workshop (ASRU)*, 2021, pp. 914–921.
- [17] A. Pasad, B. Shi, and K. Livescu, "Comparative layer-wise analysis of self-supervised speech models," in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2023, pp. 1–5.
- [18] A. Y. F. Chiu, F. K. Ching, R. Li, J. Li, and T. Lee, "A large-scale probing analysis of speaker-specific attributes in self-supervised speech representations," 2025.
- [19] J. Wagner, A. Triantafyllopoulos, H. Wierstorf, M. Schmitt, F. Burkhardt, F. Eyben, and B. W. Schuller, "Dawn of the transformer era in speech emotion recognition: Closing the valence gap," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 45, no. 9, pp. 10 745–10 759, 2023.
- [20] W. Chen, X. Xing, P. Chen, and X. Xu, "Vesper: A compact and effective pretrained model for speech emotion recognition," *IEEE Transactions on Affective Computing*, 2024.
- [21] Z. Ma, Z. Zheng, J. Ye, J. Li, Z. Gao, S. Zhang, and X. Chen, "emotion2vec: Self-supervised pre-training for speech emotion representation," in *Findings of the Association for Computational Linguistics: ACL*, 2024, pp. 15 747–15 760.
- [22] L.-W. Chen and A. I. Rudnicky, "Exploring wav2vec 2.0 fine tuning for improved speech emotion recognition," in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2023, pp. 1–5.
- [23] L. Schewski, M. Magimai Doss, G. Beldi, and S. Keller, "Measuring negative emotions and stress through acoustic correlates in speech: A systematic review," *PLOS ONE*, vol. 20, no. 7, p. e0328833, 2025.

- [24] P. Larrouy-Maestri, D. Poeppel, and M. D. Pell, “The sound of emotional prosody: Nearly 3 decades of research and future directions,” *Perspectives on Psychological Science*, vol. 20, no. 4, 2025.
- [25] F. Eyben, M. Wöllmer, and B. Schuller, “opensmile: the munich versatile and fast open-source audio feature extractor,” in *Proc. 18th ACM International Conference on Multimedia (MM)*, 2010, pp. 1459–1462.
- [26] E. Zwicker and E. Terhardt, “Analytical expressions for critical-band rate and critical bandwidth as a function of frequency,” *The Journal of the Acoustical Society of America*, vol. 68, no. 5, pp. 1523–1525, 1980.
- [27] Z. Peng, W. He, Y. Li, Y. Du, and J. Dang, “Multi-level attention-based categorical emotion recognition using modulation-filtered cochleagram,” *Applied Sciences*, vol. 13, no. 11, p. 6749, 2023.
- [28] Z. Peng, H. Zeng, Y. Li, Y. Du, and J. Dang, “Enhancing dimensional emotion recognition from speech through modulation-filtered cochleagram and parallel attention recurrent network,” *Electronics*, vol. 12, no. 22, p. 4620, 2023.
- [29] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, “Attention is all you need,” in *Advances in Neural Information Processing Systems 30 (NIPS)*, 2017, pp. 5998–6008.
- [30] R. G. Praveen, E. Granger, and P. Cardinal, “Cross attentional audio-visual fusion for dimensional emotion recognition,” in *Proc. 16th IEEE International Conference on Automatic Face and Gesture Recognition (FG)*, 2021, pp. 1–8.
- [31] H. Zou, Y. Si, C. Chen, D. Rajan, and E. S. Chng, “Speech emotion recognition with co-attention based multi-level acoustic information,” in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2022, pp. 7367–7371.
- [32] R. Zhao, X. Jiang, F. R. Yu, V. C. M. Leung, T. Wang, and S. Zhang, “Leveraging cross-attention transformer and multi-feature fusion for cross-linguistic speech emotion recognition,” 2025.
- [33] F. Li, J. Luo, and W. Xia, “WavFusion: Towards wav2vec 2.0 multimodal speech emotion recognition,” in *MultiMedia Modeling (MMM)*, ser. Lecture Notes in Computer Science, vol. 15523, 2025, pp. 325–336.
- [34] B. Nasersharif and M. Namvarpour, “Exploring the potential of wav2vec 2.0 for speech emotion recognition using classifier combination and attention-based feature fusion,” *The Journal of Supercomputing*, vol. 80, no. 16, pp. 23 667–23 688, 2024.
- [35] Y. Dong, J.-B. Cordonnier, and A. Loukas, “Attention is not all you need: pure attention loses rank doubly exponentially with depth,” in *Proc. 38th International Conference on Machine Learning (ICML)*, ser. PMLR, vol. 139, 2021, pp. 2793–2803.
- [36] S. Zhai, T. Likhomanenko, E. Littwin, D. Busbridge, J. Ramapuram, Y. Zhang, J. Gu, and J. M. Susskind, “Stabilizing transformer training by preventing attention entropy collapse,” in *Proc. 40th International Conference on Machine Learning (ICML)*, ser. PMLR, vol. 202, 2023, pp. 40 770–40 803.
- [37] A. Bibal, R. Cardon, D. Alfter, R. Wilkens, X. Wang, T. François, and P. Watrin, “Is attention explanation? an introduction to the debate,” in *Proc. 60th Annual Meeting of the Association for Computational Linguistics (ACL), Volume 1: Long Papers*, 2022, pp. 3889–3900.
- [38] S. Liu, C. Zheng, O. Demasi, S. Sabour, Y. Li, Z. Yu, Y. Jiang, and M. Huang, “Towards emotional support dialog systems,” in *Proc. 59th Annual Meeting of the Association for Computational Linguistics and the 11th IJCNLP (Volume 1: Long Papers)*, 2021, pp. 3469–3483.
- [39] Y. Qian, W. Zhang, and T. Liu, “Harnessing the power of large language models for empathetic response generation: Empirical investigations and improvements,” in *Findings of the Association for Computational Linguistics: EMNLP*, 2023, pp. 6516–6528.
- [40] S. Sabour, S. Liu, Z. Zhang, J. M. Liu, J. Zhou, A. S. Sunaryo, T. M. C. Lee, R. Mihalcea, and M. Huang, “EmoBench: Evaluating the emotional intelligence of large language models,” in *Proc. 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2024, pp. 5986–6004.
- [41] C. Tang, W. Yu, G. Sun, X. Chen, T. Tan, W. Li, L. Lu, Z. Ma, and C. Zhang, “SALMONN: Towards generic hearing abilities for large language models,” 2023.
- [42] Y. Chu, J. Xu, Q. Yang, H. Wei, X. Wei, Z. Guo, Y. Leng, Y. Lv, J. He, J. Lin, C. Zhou, and J. Zhou, “Qwen2-audio technical report,” 2024.
- [43] H. Kim, S. Seo, K. Jeong, O. Kwon, S. Kim, J. Kim, J. Lee, E. Song, M. Oh, J.-W. Ha, S. Yoon, and K. M. Yoo, “Paralinguistics-aware speech-empowered large language models for natural conversation,” in *Advances in Neural Information Processing Systems 37 (NeurIPS)*, 2024.

- [44] W. Kang, J. Jia, C. Wu, W. Zhou, E. Lakomkin, Y. Gaur, L. Sari, S. Kim, K. Li, J. Mahadeokar, and O. Kalinli, "Frozen large language models can perceive paralinguistic aspects of speech," in *Proc. Interspeech*, 2025, pp. 4323–4327.
- [45] Y. Chen, X. Xie, H. Xu, M. Geng, G. Li, C. Deng, H. Wang, S. Hu, and X. Liu, "Towards LLM-empowered fine-grained speech descriptors for explainable emotion recognition," in *Proc. Interspeech*, 2025, pp. 4633–4637.
- [46] K. Tian, E. Mitchell, A. Zhou, A. Sharma, R. Rafailov, H. Yao, C. Finn, and C. D. Manning, "Just ask for calibration: Strategies for eliciting calibrated confidence scores from language models fine-tuned with human feedback," in *Proc. 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023, pp. 5433–5442.
- [47] B. Wen, J. Yao, S. Feng, C. Xu, Y. Tsvetkov, B. Howe, and L. L. Wang, "Know your limits: A survey of abstention in large language models," *Transactions of the Association for Computational Linguistics*, vol. 13, pp. 529–556, 2025.
- [48] S. Kapoor and A. Narayanan, "Leakage and the reproducibility crisis in machine-learning-based science," *Patterns*, vol. 4, no. 9, p. 100804, 2023.
- [49] A. Elangovan, J. He, and K. Verspoor, "Memorization vs. generalization: Quantifying data leakage in NLP performance evaluation," in *Proc. 16th Conference of the European Chapter of the ACL (EACL): Main Volume*, 2021, pp. 1325–1335.
- [50] M. Rosenblatt, L. Tejavibulya, R. Jiang, S. Noble, and D. Scheinost, "Data leakage inflates prediction performance in connectome-based machine learning models," *Nature Communications*, vol. 15, p. 1829, 2024.
- [51] K. Gorman and S. Bedrick, "We need to talk about standard splits," in *Proc. 57th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2019, pp. 2786–2791.
- [52] X. Bouthillier, P. Delaunay, M. Bronzi, A. Trofimov, B. Nichyporuk, J. Szeto, N. Mohammadi Sepahvand, E. Raff, K. Madan, V. Voleti, S. Ebrahimi Kahou, V. Michalski, T. Arbel, C. Pal, G. Varoquaux, and P. Vincent, "Accounting for variance in machine learning benchmarks," in *Proceedings of Machine Learning and Systems 3 (MLSys)*, 2021.
- [53] M. A. Lones, "Avoiding common machine learning pitfalls," *Patterns*, vol. 5, no. 10, p. 101046, 2024.
- [54] B. Schuller, S. Steidl, and A. Batliner, "The INTERSPEECH 2009 emotion challenge," in *Proc. Interspeech*, 2009, pp. 312–315.
- [55] A. R. Naini, L. Goncalves, A. N. Salman, P. Mote, I. R. Ulgen, T. Thebaud, L. Moro Velazquez, L. P. Garcia, N. Dehak, B. Sisman, and C. Busso, "The Interspeech 2025 challenge on speech emotion recognition in naturalistic conditions," in *Proc. Interspeech*, 2025, pp. 4668–4672.
- [56] Z. Ma, M. Chen, H. Zhang, Z. Zheng, W. Chen, X. Li, J. Ye, X. Chen, and T. Hain, "EmoBox: Multilingual multi-corpus speech emotion recognition toolkit and benchmark," in *Proc. Interspeech*, 2024, pp. 1580–1584.
- [57] E. Ibrahim, M. E. Ghoraba, and A. E. Ghoraba, "Multimodal emotion recognition using hybrid deep feature fusion under speaker-independent evaluation," *Scientific Reports*, vol. 16, p. 19584, 2026.
- [58] C. Luna-Jiménez, R. Kleinlein, D. Griol, Z. Callejas, J. M. Montero, and F. Fernández-Martínez, "A proposal for multimodal emotion recognition using aural transformers and action units on RAVDESS dataset," *Applied Sciences*, vol. 12, no. 1, p. 327, 2022.
- [59] F. Portal, J. de Lope, and M. Graña, "A performance benchmarking review of transformers for speaker-independent speech emotion recognition," *International Journal of Neural Systems*, vol. 35, no. 10, p. 2530001, 2025.
- [60] J. M. Hoenig and D. M. Heisey, "The abuse of power: The pervasive fallacy of power calculations for data analysis," *The American Statistician*, vol. 55, no. 1, pp. 19–24, 2001.
- [61] International Organization for Standardization, "ISO 532-1:2017 acoustics — methods for calculating loudness — part 1: Zwicker method," ISO, Geneva, 2017.
- [62] J. L. Ba, J. R. Kiros, and G. E. Hinton, "Layer normalization," 2016.
- [63] D. Hendrycks and K. Gimpel, "Gaussian error linear units (GELUs)," 2016.
- [64] N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov, "Dropout: A simple way to prevent neural networks from overfitting," *Journal of Machine Learning Research*, vol. 15, no. 56, pp. 1929–1958, 2014.
- [65] M. Abdin, J. Aneja, H. Awadalla, A. Awadallah, A. A. Awan, N. Bach, A. Bahree, A. Bakhtiari, J. Bao, H. Behl *et al.*, "Phi-3 technical report: A highly capable language model locally on your phone," 2024.
- [66] R. Agarwal, M. Schwarzer, P. S. Castro, A. C. Courville, and M. G. Bellemare, "Deep reinforcement learning at the edge of the statistical precipice," in *Advances in Neural Information Processing Systems 34 (NeurIPS)*, 2021.

- [67] J. Demšar, “Statistical comparisons of classifiers over multiple data sets,” *Journal of Machine Learning Research*, vol. 7, pp. 1–30, 2006.
- [68] T. G. Dietterich, “Approximate statistical tests for comparing supervised classification learning algorithms,” *Neural Computation*, vol. 10, no. 7, pp. 1895–1923, 1998.
- [69] D. Card, P. Henderson, U. Khandelwal, R. Jia, K. Mahowald, and D. Jurafsky, “With little power comes great responsibility,” in *Proc. 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2020, pp. 9263–9274.
- [70] I. Loshchilov and F. Hutter, “Decoupled weight decay regularization,” in *7th International Conference on Learning Representations (ICLR)*, 2019.
- [71] —, “SGDR: Stochastic gradient descent with warm restarts,” in *5th International Conference on Learning Representations (ICLR)*, 2017.
- [72] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, and Z. Wojna, “Rethinking the inception architecture for computer vision,” in *Proc. IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 2818–2826.
- [73] T. Leygue, A. Sabourin, C. Bolzmacher, S. Bouchigny, M. Anastassova, and Q.-C. Pham, “Explainable speech emotion recognition through attentive pooling: Insights from attention-based temporal localization,” 2025.
- [74] H. Cao, D. G. Cooper, M. K. Keutmann, R. C. Gur, A. Nenkova, and R. Verma, “CREMA-D: Crowd-sourced emotional multimodal actors dataset,” *IEEE Transactions on Affective Computing*, vol. 5, no. 4, pp. 377–390, 2014.
- [75] F. Burkhardt, A. Paeschke, M. Rolfes, W. F. Sendlmeier, and B. Weiss, “A database of German emotional speech,” in *Proc. Interspeech*, 2005, pp. 1517–1520.
- [76] C. Busso, M. Bulut, C.-C. Lee, A. Kazemzadeh, E. Mower, S. Kim, J. N. Chang, S. Lee, and S. S. Narayanan, “IEMOCAP: interactive emotional dyadic motion capture database,” *Language Resources and Evaluation*, vol. 42, no. 4, pp. 335–359, 2008.